CT-01

TOPIC NAME : DIP DAY : TIME : DATE : / /

Slide-01

⊛ why dig image processing is necessary ?

→ Digital images are necessary because they transform visual information into manipulable data or changeable data that computers can be instantly understand, stored, edited and shared globally.

Digital images focuses on two major tasks —

① Improvement of pictorial or image information for human interpretation (such as, X-ray, CT-scan report)

② Making or processing of image data for store, send or transmission and represent or understandable for autonomous machine or computers perception (মেশিনের) or insight or understanding (such as, self drive car, face unlock)

CT-01

⊛ what is digital image ?

→ A digital image is a ~~finite~~ representation of a 2D visual scene, defined by a discrete function $f(x, y)$

x, y ⇒ coordinates spatial (স্থানিক) location of a point in image.

f ⇒ (amplitude) Its intensity or gray level at specific point

pixels ⇒ smallest individual elements of digital image. that containing both specific location & intensity or gray level value.

GOOD LUCK

TOPIC NAME : _____ DAY : _____

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⊛ what is digital Image processing?

→ Digital Image processing is the use of computer algorithms to enhance on extract information from digital images.

It's ~~occur~~ categorize in 3-levels

(i) Low level (input (image) to output (image))

(ii) Mid level (input (image) to output (image attribute))

(iii) High level (input (image attribute) to output (image understand))

⊛ History of DIP?

(i) 1920s (Beginning) → The Bartlane cable picture transmission system allowed images to be sent across the Atlantic via submarine cables, reducing transmission time from weeks to hours. (1st app was newspaper digital imaging). late 1920 Bartlane improve high quality image based on photographic techniques & increased number of tones.

(ii) 1960s → Space Race

modern DIP was born at NASA's Jet propulsion ~~laboratory~~ laboratory (JPL) to improve images sent back from Ranger 7.

(iii) 1970s → Medical Revolution

Invention of (CAT scans) computerized Axial Tomography scans used DIP to create 3D views of human body.

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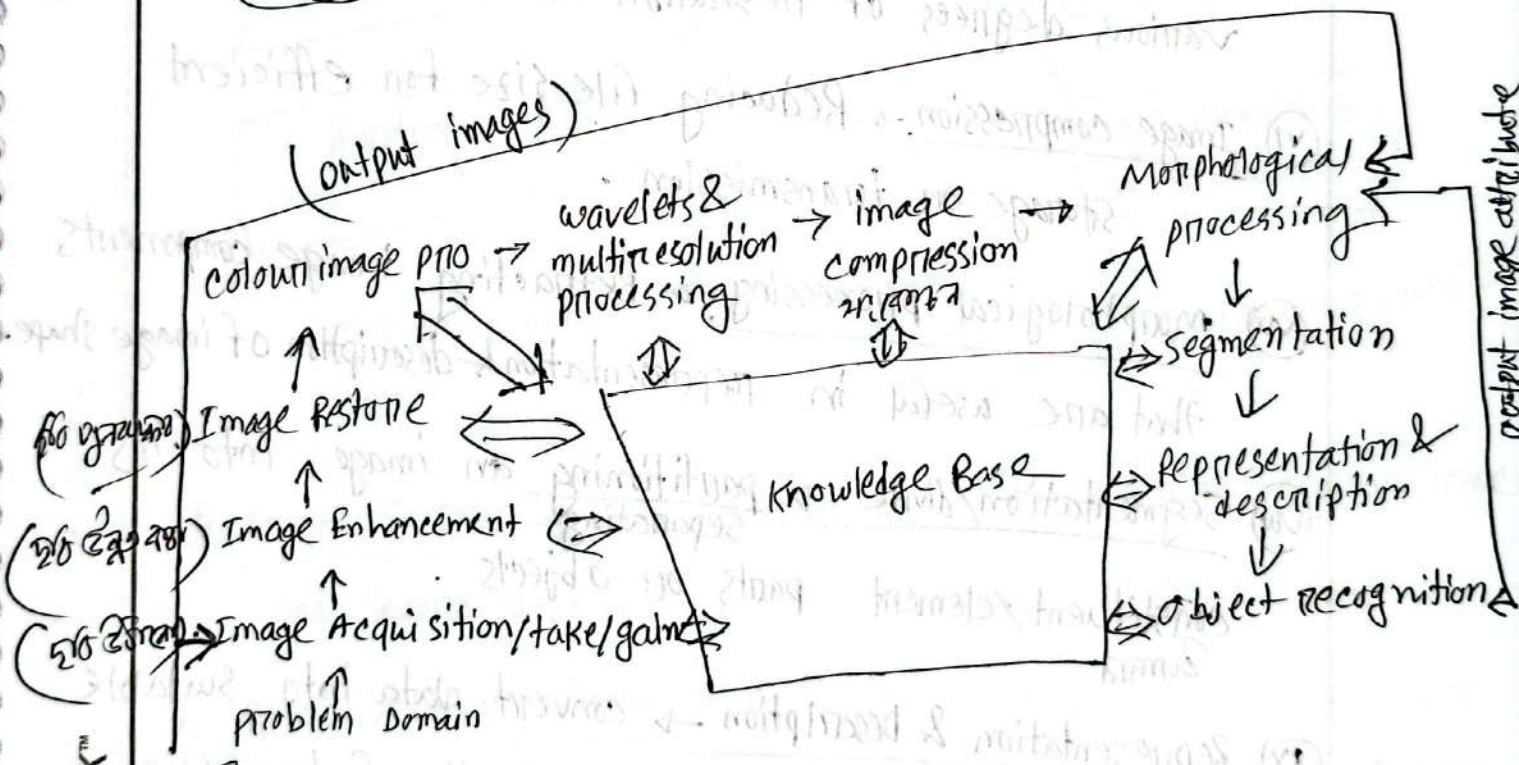
TIME : DATE : / /

1950s – today → Modern Era

DIP expanded into digital cameras, satellite imaging, AI. for facial recognition & autonomous driving.

⑧ Fundamental Steps in dig IP ?

(10 steps) ⇒ (capturing, cleaning, analyzing & recognizing images)



Steps

① Image Acquisition/capture → image captured by camera or sensor. convert image using analog to digital converter.

② Image enhancement → Improving visual quality of image. (Brightness, contrast adjust)

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(III) Image Restore → using mathematical models to fix degradations/degeneration, such as remove noise/blur.

(IV) Colour Image processing → Handle color modeling and processing in a digital domain.

(V) Wavelets & multiresolution processing → Represent images in various degrees of resolution for data compression (압축)

(VI) Image compression → Reducing file size for efficient storage or transmission.

(VII) Morphological processing → Extracting image components that are useful in representation & description of image shape.

(VIII) Segmentation/divide → partitioning/separating an image into its constituent/element parts or objects.
2ormra

(IX) Representation & Description → convert data into suitable forms for representation and extracting features like size or shape for description.

(X) Object Recognition → Assign level based on descriptor or definition.

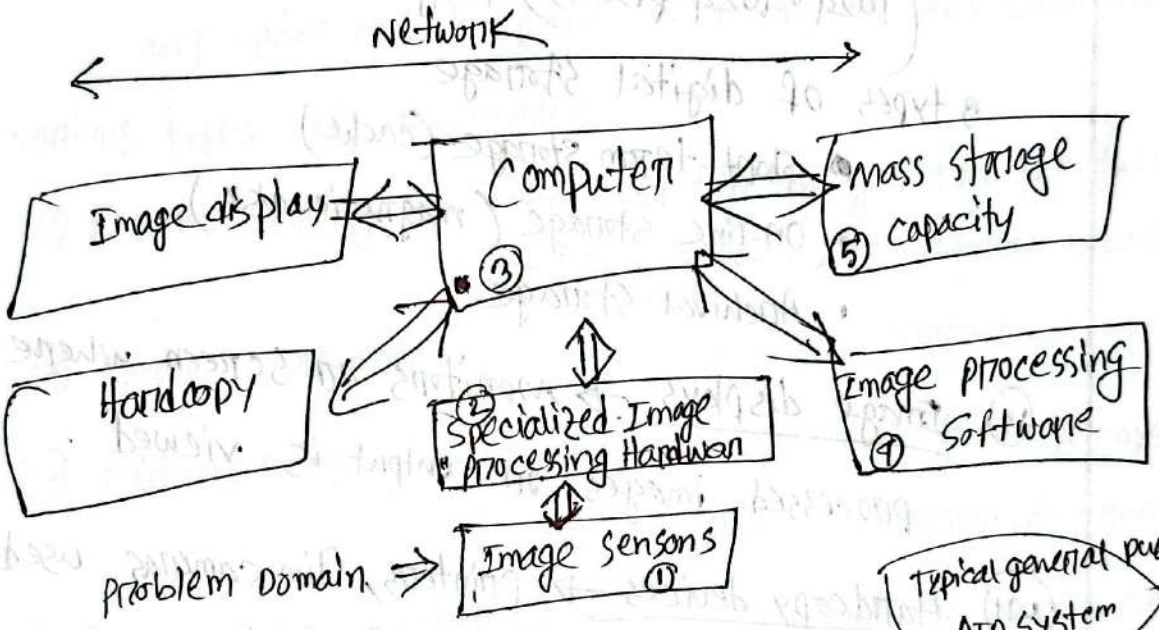
"GOOD LUCK"

TOPIC NAME : Slide-02 DAY : _____

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(*) Components of Image processing System ?

[Systems consists of capture tools as sensors, processing power as computer and output/storage as display/networks]



(i) Image sensors → first physical device (camera sensor) capture image and second (digitizer) convert light into electric signal.

(ii) Specialized IP Hardware → Dedicated hardware like GPU or DSP that performs high-speed mathematical operations (ALU) on image data.

(iii) Computer → CPU that manages entire system and runs general tasks.

(iv) Image processing software → Specialized programs and algorithms like MATLAB that perform specific tasks used to manipulate the image.

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(v) Mass storage capacity → (HDD or cloud storage) that used to save large image file for future use.

(1024*1024 pixels) image size

3 types of digital storage

- Short-term storage (cache)
- On-line storage (magnetic disk)
- Archival storage

(vi) Image displays → monitors or screen where the processed images or output is viewed.

(vii) Hardcopy devices → printers, film cameras used to produce a physical copy of the images.

(viii) Networking → connection that allows the system to transmit and share image data with other device.

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RBMBS

⑧ Application of digital Image processing ?

[DIP is used whenever a human or machine needs to visualize, transmit, or understand 2D data from deep space and hospitals to factories and security]

various types of applications

(i) Remote sensing → using satellite images for weather forecasting (पूर्वानुमान), agricultural tracking.

(ii) Space exploration → Analyze image to recognize

(iii) Medical Imaging → processing images report of X-rays, MRI, CT-scan for patient diagnosis & treatment.

(iv) Business & communication → used in broadcast TV, video conference and CCTV security systems.

(v) Radar & Sonar → Detecting and recognizing targets for aircraft navigation (नेविगेशन) or missile guidance systems.

(vi) Robotics & Industry → Enable robot vision for automated factory inspection of industrial parts.

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⑧ Image formation in eye?

monochromatic → colourless or void of color

Chromatic light → colored light that electromagnetic energy spectrum from 0.43 to 0.79 um.

Radiance → Total energy from the source.

Luminance (উজ্জ্বলতা) → Energy perceived (অনুভূত শক্তি) by observer.

Brightness → Subjective feeling that can't be measured.

$f(x,y) = i(x,y) \times r(x,y)$ → intensity of any pixel

⑧ An image is denoted by a func $f(x,y)$, which f at spatial coordinates (x,y) is a scalar quantity proportional to energy radiated by a physical source.

- The values of f are non-negative and finite $0 \le f(x,y) < inf$
- func $f(x,y)$ is characterized by 2 components.

① Illumination (আলোকিতকরণ) → $[i(x,y)]$ The amount of light falling on the object from a source.

② Reflectance (প্রতিফলন) → $[r(x,y)]$ (0 to 1) The amount of that light reflected back by the object.

$f(x,y) = i(x,y) * r(x,y)$ ← (intensity of pixels)

$0 \le i(x,y) < inf$

$0 \le r(x,y) \le 1$

$f(x,y)$ = 0 হলে শুধু কৃষ্ণবর্ণ বোঝাবে.

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Sampling → Digitizes the spatial coordinates on pixels (x, y) → coordinate value

Quantization → Digitizes the intensity levels (f) to create a digital image. → Amplitude value

Uniform Sampling

- Digitized in spatial domain ($I_{M \times N}$)
- M & N are usually integer powers of two
- Nyquist theorem (captured image accurately) and Aliasing (visual distortion)
  - → $f_s \geq 2 \times f_{max}$
  - → $f_s < 2 \times f_{max}$

Pixels

No. of pixels = N * M

No. of bits = N * M * K

N * M = all pixels num

K = no. of bits in each pixel

L = grayscale levels = 2^K

① pixels — 128 × 128 and 8 bit, Grayscale level = ?

Grayscale level, $L = 2^K = 2^8 = 256$ level ($N = M = 128$)

Total bits = $N \times M \times K$

$= 128 \times 128 \times 8$ bits

$= \frac{128 \times 128 \times 8}{8}$ byte

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(*) Some relationship between pixels?

(i) Neighbors of a pixels

- 4-neighbors $[N_4(p)]$: 4 pixels (Top, bottom, left, right)

coordinates $(x+1, y), (x-1, y), (x, y+1), (x, y-1)$

(ii) Diagonal neighbors of pixels $[N_D(p)]$:

coordinates $(x+1, y+1)(x-1, y+1), (x+1, y-1), (x-1, y-1)$

(iii) 8-neighbors $[N_8(p)]$:

4-neighbors + Diagonal neighbors

(2) Adjacency

(3) connectivity

(4) Distance measures

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(*) Arithmatic Operation → two type of Operation in DIP

Logical → AND, OR, NOT

Arithmatic

Addition
Subtraction
Multiplication
Division

(*) $f(x,y)$ & $g(x,y)$

$s(x,y) = f(x,y) + g(x,y)$ (add)

$d(x,y) = f(x,y) - g(x,y)$ (sub)

$M(x,y) = f(x,y) * g(x,y)$ (mul)

$v(x,y) = f(x,y) / g(x,y)$ (Div)

(*) represent image:

$$\begin{bmatrix} 0 & 100 & 10 \\ 4 & 0 & 10 \\ 5 & 6 & 7 \end{bmatrix} + \begin{bmatrix} 255 & 0 & 108 \\ 5 & 5 & 155 \\ 9 & 13 & 15 \end{bmatrix} = \begin{bmatrix} 255 & 100 & 20 \\ 9 & 5 & 165 \\ 14 & 19 & 22 \end{bmatrix}$$

pixel range of image is = [0 - 255]

① if pixel is (value) is greater than 255, so it will be 255 not big value

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(ii) if pixel value is infinite then it will be replace by zero (0).

(iii) if pixel value is negative, then it will be minimum pixel value 0 or others [0-255] → minimum = 0

[10-255] → " = 10

(iv) if pixel value is float then it will be floor or ceil

$$\begin{bmatrix} 0 & 100 & 10 \\ 9 & 0 & 10 \\ 5 & 6 & 7 \end{bmatrix} - \begin{bmatrix} 255 & 0 & 10 \end{bmatrix} = \begin{bmatrix} & & \end{bmatrix}$$

(v) Logical Operation use in Binary Image

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} \text{AND} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

↓ NOT ↳ OR = $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$

$$\begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

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point Operation

↳ output pixel value depends on its input pixel's value.

$S = T(r)$ where, r = gray value of input image, S = " " of output "

Image Negative method / Digital Negative:

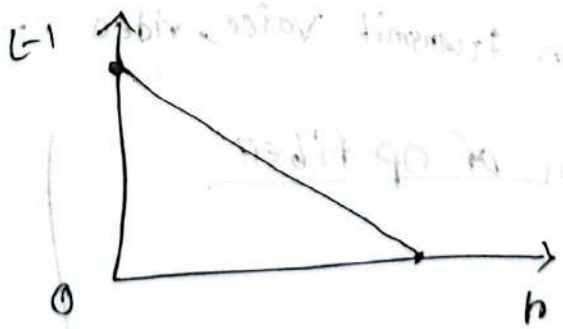
$S = (L-1) - r$ where, L = max range of input image bits

input image

$$\begin{bmatrix} 4 & 3 & 5 & 2 \\ 3 & 6 & 4 & 6 \\ 2 & 2 & 6 & 5 \\ 7 & 6 & 4 & 1 \end{bmatrix} = \begin{bmatrix} (L-1)-h \text{ or } (8-1)-3 & 4 & 2 & 5 \\ 4 & 1 & 3 & 1 \\ 5 & 5 & 1 & 2 \\ 0 & 1 & 3 & 6 \end{bmatrix}$$

$\therefore L-1 = 8-1 = 7$

$2^3 = 8$ bit মানে represent করতে $L = 8$ bits



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TOPIC NAME : [explanation important]

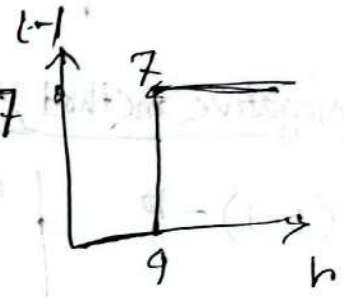
DAY : Saturday

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Thresholding method :

$$S_2 = \begin{cases} L-1 & \text{if } b \geq q \\ 0 & \text{if } b < q \end{cases}$$

$$S_2 = \begin{bmatrix} 7 & 0 & 7 & 0 \\ 0 & 7 & 7 & 7 \\ 0 & 0 & 7 & 7 \\ 7 & 7 & 7 & 0 \end{bmatrix} \rightarrow$$



OC - 04.05.26

Sampling → Nyquist theorem

$N_s \geq 2 \times 100 \quad (f_B = 100)$

why OP fiber preferred?

(i) High BW (ii) Long distance (iii) Immunity to electromagnetic interference

(iv) It can transmit voice, video ...

Application of OP fiber

Signal → frequency

light → wavelength

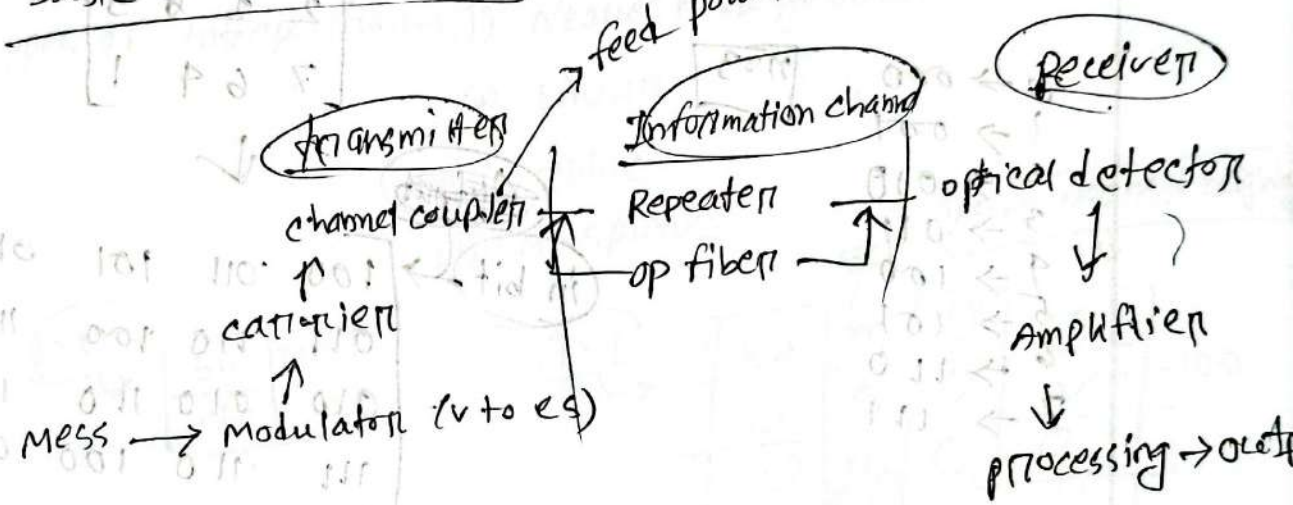
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Scattering → বাধা অথবা সমস্যার জন্য error spread (ছড়িয়ে পড়া).

Basic Block dia of OF



DIP — 04.05.26

(14) Q. Clipping method

$$S = \begin{cases} L-1 & - \; 2 \le r \le 5 \\ 0 & - \; \text{otherwise} \end{cases}$$

$L = 8$

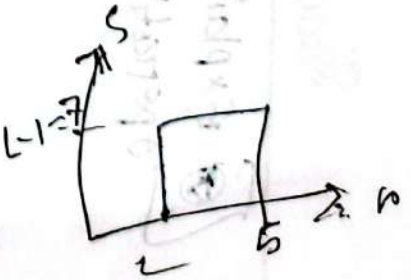
$(L-1 = 7)$

input image :

$$\begin{bmatrix} 4 & 3 & 5 & 2 \\ 3 & 6 & 4 & 6 \\ 2 & 2 & 6 & 5 \\ 7 & 6 & 4 & 1 \end{bmatrix} \rightarrow$$

output

$$\begin{bmatrix} 7 & 7 & 7 & 7 \\ 7 & 0 & 7 & 0 \\ 7 & 7 & 0 & 7 \\ 0 & 0 & 7 & 0 \end{bmatrix}$$



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(iv) Bit-plane slicing

input →
$$\begin{bmatrix} 4 & 3 & 5 & 2 \\ 3 & 6 & 4 & 6 \\ 2 & 2 & 6 & 5 \\ 7 & 6 & 4 & 1 \end{bmatrix}$$

$L = 8 = 2^n$

$n = 3$

$= 2^3$

$n = 3$

0 → 000
1 → 001
2 → 010
3 → 011
4 → 100
5 → 101
6 → 110
7 → 111

in bit →
$$\begin{bmatrix} 100 & 011 & 101 & 010 \\ 011 & 110 & 100 & 110 \\ 010 & 010 & 110 & 101 \\ 111 & 110 & 100 & 001 \end{bmatrix}$$

Ques

Explain the bit-plane Slicing operation for DIP

most MSB →
$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

middle SB

Lowest SB →
$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

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Interpolation :

Types of interpolation → (i) Nearest neighbour
(ii) Linear
(iii) cubic
(iv) B-spline

2×2 →

50	100
40	10

4×4 →

50	m	0	100
0	0	0	v
0	n	0	0
40		w	10

Using nearest neighbour

50 → 100 → 40 → 10

Given

(*) calculate the pixel value using Linear & cubic interpolation

Geometric Spatial Transformation

Techniques

(i) Rigid
- Translation
- Rotation

(ii) Non-Rigid
- Affine
- perspective
- Elastic

Table 2.2 → Matrix

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TOPIC NAME : DIP – 05.05.26 DAY: TIME: DATE: / /

Ques: Relationship between Neighbouring pixel and distance measuring pixel)
d4, d8, mdis

Arithmatic & logical

DIP – 10.05.26 (Chap–03)

Image Enhancement

- Spatial domain (involves directly pixel operations for intensity)
- Frequency domain
- combination

Basic Intensity transformation	piecewise Linear intensity transformation
• Image negative	• Contrast streteling
• Log transformation	• Intensity level slicing
• power law or gamma transformation	• Bit-plane slicing
• Thresholding	

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* Explain all method with their equation corresponding their example & graph
(ques) (also read) ↳ application of their and practice math,

TOPIC NAME: _____ DAY: TIME: DATE:

* Log Transformation method :

eqn: $s = c \log_{10}(1+r)$; $r \geq 0$

$S = (L-1) - r$

r = pixel value of II

S = " " of OI

L = max value of II

: consider value ca

$c = 1$

$c = \frac{L}{\log_{10}(L+1)}$

Input Image:

110	120	90
91	94	98
90	91	99

$2^t = 128$

$L = 128 - 1 = 127$

for 1st case:

$c = 1$

$S = 1 \times \log_{10}(1 + 110)$ (r)

$= 2.04$

output Image:

2.04	2.08	1.95
1.96	1.97	1.99
1.95	1.96	2

after rounding:

$\approx$

2	2	2
2	2	2
2	2	2

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For 2nd case

$$c = \frac{L}{\log_{10}(L+1)} \quad (L = 127)$$

$$= \frac{127}{\log_{10} 126} \approx 60.46$$

output Image:

$$S = 60.46 * \begin{bmatrix} 2.04 & 2.08 & 1.95 \\ 1.96 & 1.97 & 1.99 \\ 1.95 & 1.96 & 2 \end{bmatrix}$$

(*) power law or gamma Transformation method:

eqn: $S = cr^{\gamma}$; $\gamma > 1$ (increase), $\gamma < 1$ (decrease)

c, γ = constant

r = input image pixel value

1st case:

c = 1, γ = 0.5

$S = 1.r^{\gamma}$

$= 110^{0.5}$

output Imge:

$$\begin{bmatrix} 10.48 & 10.95 & 9.48 \\ 9.53 & 9.69 & 9.89 \\ 9.48 & 9.53 & 9.94 \end{bmatrix}$$

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TOPIC NAME : _____ DAY : _____

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2nd case :

$c=1, \gamma=1.5$

output Image

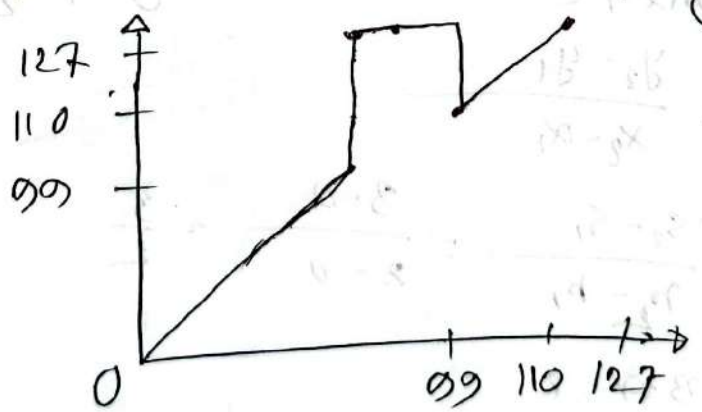
$$S = \begin{bmatrix} 127 & 127 & 127 \\ 127 & 127 & 127 \\ 127 & 127 & 127 \end{bmatrix}$$

[if output pixel value is greater than L, so replace these whole value to maximum pixel value (L=127)]

Intensity level slicing method :

Eg: (i) with Background → clipping method $\left[S = \begin{cases} L-1 & 99 \le n \le 110 \\ 0 & \text{otherwise} \end{cases}\right]$

(ii) without " → n $\left[S = \begin{cases} L-1, & 99 \le n \le 110 \\ n, & \text{otherwise} \end{cases}\right]$



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consider input image

$$\begin{bmatrix} 4 & 7 & 5 & 6 \\ 2 & 3 & 1 & 0 \\ 4 & 3 & 5 & 7 \end{bmatrix}$$

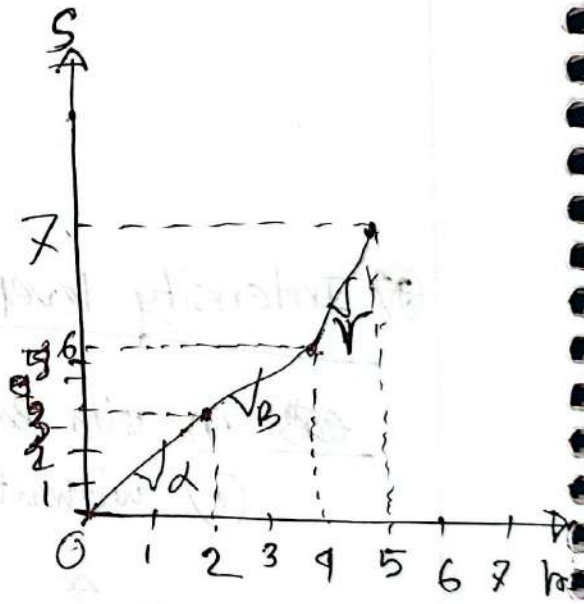
(i) with Bg $\quad s = \begin{cases} L-1, & (\quad \le r \le \quad) \\ 0, & \text{otherwise} \end{cases}$

(ii) without Bg $\quad s = \begin{cases} L-1, & (\quad \le r \le \quad) \\ r, & \text{otherwise} \end{cases}$

(*) Contrast stretching method :

eqn:

$$s = \begin{cases} r=2, & s=3 \\ r=5, & s=5 \end{cases}$$



$$y = mx + c$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\alpha = \frac{s_2 - s_1}{r_2 - r_1} = \frac{3-0}{2-0} = \frac{3}{2}$$

Same, $\beta = \gamma = ?$

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$$S = \begin{cases} a = n & 0 \le n < 2 \\ \beta \cdot (n - n_1) + s_1, & 2 \le n < 4 \\ \gamma \cdot (n - n_2) + s_2, & 4 \le n < 7 \end{cases}$$

eqn of CSM

~~Ques: Applica~~

CC & L → 10.05.26

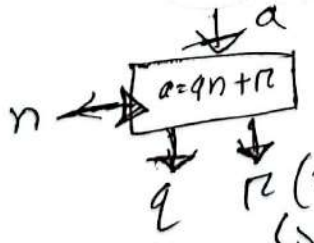
$21x + 14y = 35$

particular solution
general solution

Example → 2-13 (28 page)

2-2 → modular arithmatic
↳ ($a \bmod m = n$)

$a = qm + r$



Ques:
diff btn remainder & residues.

~~-18 | 14~~ |

141 | -18 | 2
+28

710

TOPIC NAME : _____ DAY : _____

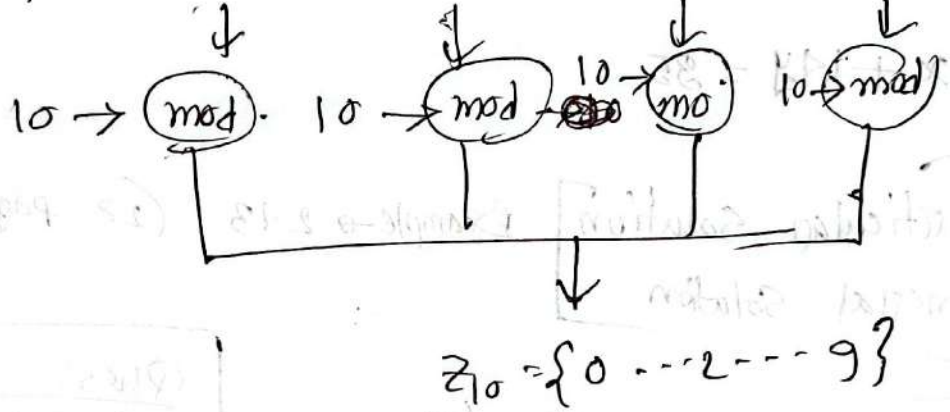
TIME : DATE : / /

31-7 | 3
9
+2

(30 page)

(*) what are the diff btn equility & congruance

$\mathbb{Z} = \{ \ldots -8 \ldots 2 \ldots 12 \ldots 22 \ldots \}$



$\mathbb{Z}_{10} = \{0 \ldots 2 \ldots 9\}$

H.W → 2.16 Example
2.17 2.18 2.19 2.20

GOOD LUCK

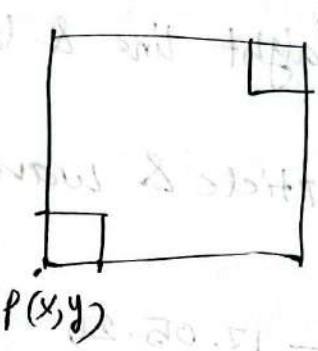
DIP Assignment - 02

TOPIC NAME: DIP-CT-02 DAY: _____

TIME: DATE: / /

Distance measures b/n pixels

Let p & q be pixels with co-ordinates (x,y) and (s,t) respectively.



(i) Euclidean distance:

$$D_E(p,q) = [(x-s)^2 + (y-t)^2]^{1/2}$$

(ii) City Block distance

$$D_4(p,q) = |x-s| + |y-t|$$

(Ex) The pixels with D_4 distance ≤ 2 from (x,y) from the following counters of constant distance.

```
        3
      3 2 3
    3 2 1 2 3
  3 2 1 0 1 2 3
    3 2 1 2 3
      3 2 3
        3
```

[it can't travel diagonally]

[0 → centre point]

4-Adjacency

GOOD LUCK

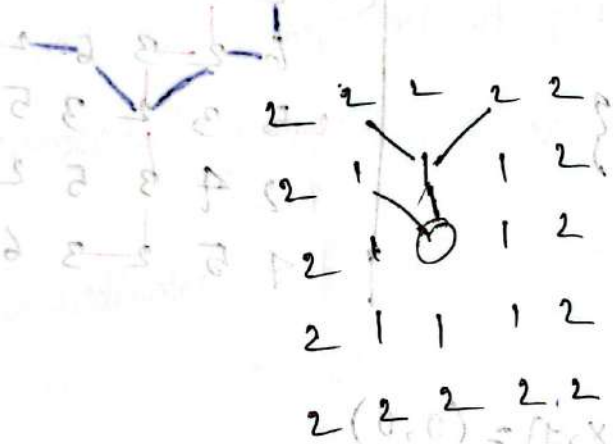
TOPIC NAME : _____ DAY : _____

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(iii) **chessboard distance:**

$$D_8(p,q) = \max(|x-s|, |y-t|)$$

(Ex) → The pixels with D_8 distance ≤ 2 from (x,y) form the following counters of constant distance:



(it can go diagonally also)

(0 → centre)

(8-Adjacency)

(iv) **D_M distance:**

It is defined as the shortest m-path btn the points. This distance btn 2 pixels depends on the values of the pixels along the path as well as the values of their neighbours.

(2-points) $p(x,y)$ ——— $q(s,t)$ (co-ordinates)

(Note) → The D_4 & D_8 distance btn p & q are independent of any path that might exist btn the points because these distance involve only the co-ordinates of the points.

GOOD LUCK

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Question:

① An image segment is shown below. Let V be the set of gray level values used to define connectivity in the image. Compute D_4, D_8 and D_M distance btn pixels 'p' and 'q' for:

① $V = \{2, 3\}$

② $V = \{2, 6\}$

p(x,y)	1	2	3	4
0	2	3	2	6
1	6	2	3	6
2	5	3	2	3
3	2	4	3	5
4	4	5	2	3

(0,0) = p(x,y) ; (4,4) = q(s,t)

Soln :

co-ordinates of $p(x,y) = (0,0)$

" of $q(s,t) = (4,4)$

$D_4(p,q) = |x-s| + |y-t| = |0-4| + |0-4| = 8$ units

$D_8(p,q) = \max(|x-s|, |y-t|) = \max(|0-4|, |0-4|) = 4$ "

① $V = \{2, 3\}$ (Reach p to q) (if reached OK otherwise no path)

$D_M(p,q)$ =

↳ It's depends on path (D_M)

There is no path btn p and q is not included in the set V.

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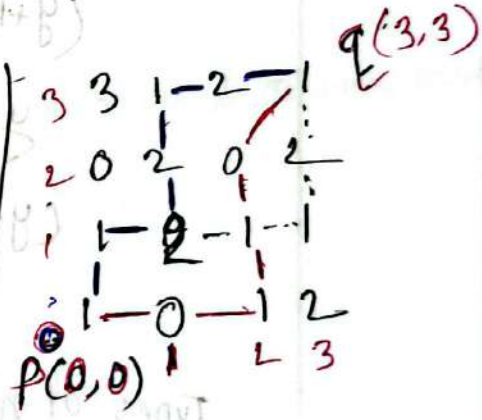
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(ii) $V = \{2, 6\}$ [$V = \{2, 6\}$]

There is no path btn p and q.

Question-2:

consider the following image segment. Compute D_M, D_4 and D_8 distances btn pixels p and q for $V = \{0, 1\}$ where V is the set of gray level values used to define connectivity. Repeat for $V = \{1, 2\}$.



Soln:

co-ordinates of $p(x,y) = (0,0)$

" of $q(s,t) = (3,3)$

$D_4(p,q) = |x-s| + |y-t| = |0-3| + |0-3|$

$= 6$ units

$D_8(p,q) = \max(|x-s|, |y-t|)$

$= \max(|0-3|, |0-3|) = \max(3,3) = 3$ units

(i) $V = \{0, 1\}$

$D_M = 5$ units (reach p to q)

(ii) $V = \{1, 2\}$

$D_M = 6$ units (shortest path)

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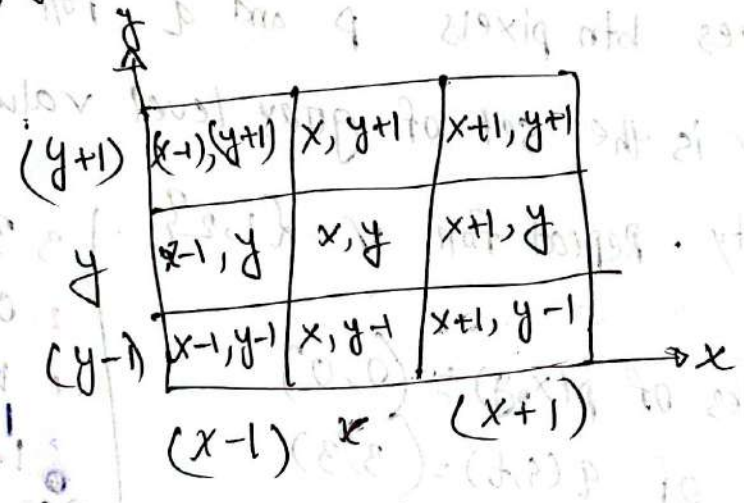
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Relationship btn pixels (Neighbourhood and Adjacency)

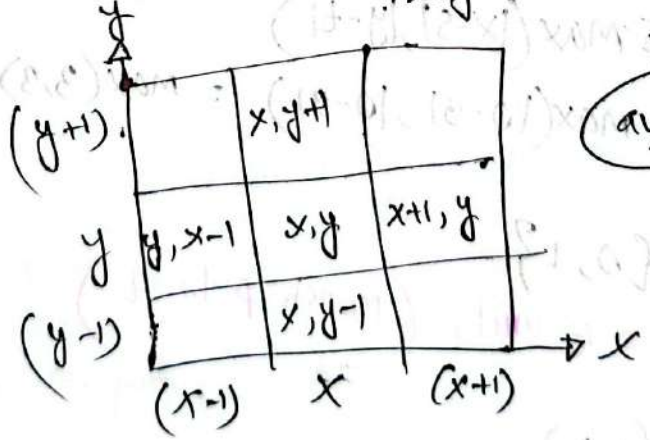
① Neighbourhood of pixels

Any image can be represented as follow :



Types of neighbourhood

① $N_4(P)$: 4-neighbours → set of horizontal and vertical neighbours.



avoid diagonal

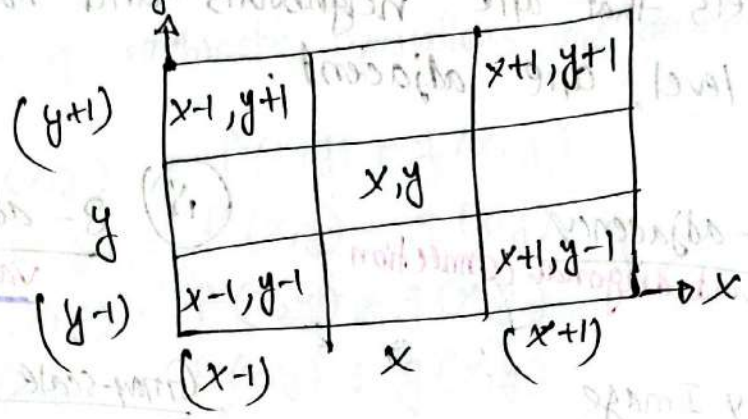
(diag to diag) shown 2 = m1

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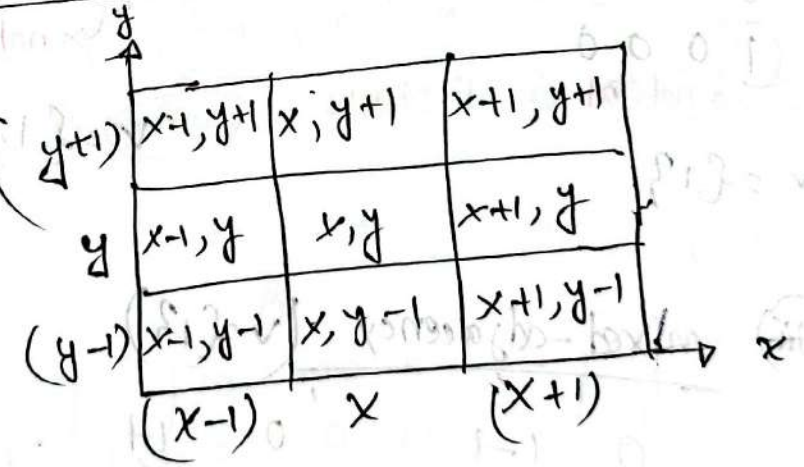
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(ii) $N_D(p)$ diagonal neighbours: Set of 4 diagonal neighbours



(iii) $N_8(p)$ 8 neighbours: union of 4 neighbours & diagonal neigh

$$N_8(p) = N_4(p) + N_D(p)$$



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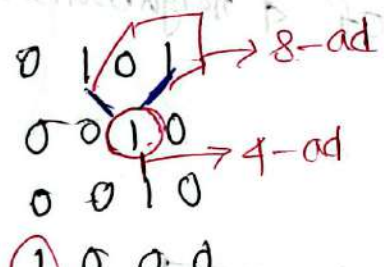
(2) Connectivity / Adjacency

2 pixels that are neighbours and have the same gray level are adjacent.

(i) 4-adjacency

avoid diagonal connection

Binary Image

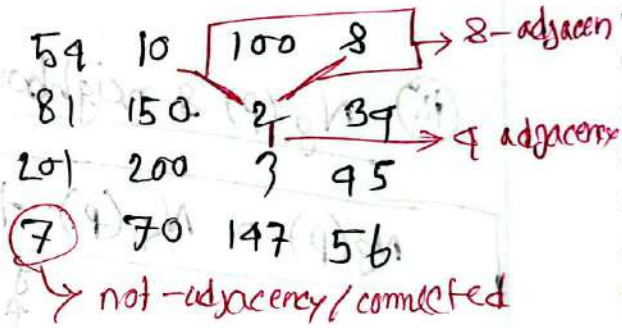


$V = \{1\}$

(ii) 8-adjacency

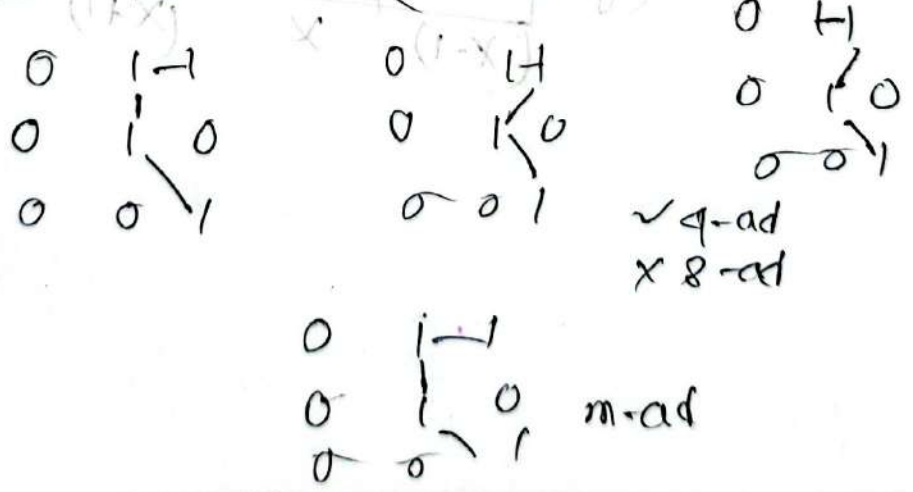
valid diagonal conn

Gray-scale Image



$V = \{1, 2, 3, \dots 10\}$

(iii) mixed-adjacency ($V = \{1\}$)



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Arithmetic Operations btn Images

These are array eqn which carried out btn corresponding pixel pairs. The 4 arithmetic operations are denoted as:

$$Sum(x,y) = f(x,y) + g(x,y)$$
$$Sub(x,y) = f(x,y) - g(x,y)$$
$$Mul(x,y) = f(x,y) * g(x,y)$$
$$Div(x,y) = f(x,y) \div g(x,y)$$

Notes:

- If result is float point num, round off it's value.
- If result is above pixel range, select max range it's value
- If result is below " " " min " " "
- If " is infinity, write it is as zero.

0-255

Addition

$$\underset{A}{\begin{bmatrix} 0 & 100 & 10 \\ 4 & 0 & 10 \\ 8 & 0 & 5 \end{bmatrix}} + \underset{B}{\begin{bmatrix} 10 & 100 & 5 \\ 2 & 0 & 0 \\ 0 & 10 & 10 \end{bmatrix}} = \begin{bmatrix} 10 & 200 & 15 \\ 6 & 0 & 10 \\ 8 & 10 & 15 \end{bmatrix}$$

Uses:

- Addition of noisy images for noise reduction.
- Image averaging in the field of astronomy.

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Subtraction

0-255

$$\underset{A}{\begin{bmatrix} 0 & 100 & 10 \\ 4 & 0 & 10 \\ 8 & 0 & 5 \end{bmatrix}} - \underset{B}{\begin{bmatrix} 10 & 100 & 5 \\ 2 & 0 & 0 \\ 0 & 10 & 10 \end{bmatrix}} = \begin{bmatrix} 0 & 0 & 5 \\ 2 & 0 & 10 \\ 8 & 0 & 0 \end{bmatrix}$$

uses :

- Enhancement of diff btn range.
- Mask mode radiography in medical Imaging.

Multiplication

0-255

$$\underset{A}{\begin{bmatrix} 0 & 100 & 10 \\ 4 & 0 & 10 \\ 8 & 0 & 5 \end{bmatrix}} * \underset{B}{\begin{bmatrix} 10 & 100 & 5 \\ 2 & 0 & 0 \\ 0 & 10 & 10 \end{bmatrix}} = \begin{bmatrix} 10 & 255 & 50 \\ 8 & 0 & 0 \\ 0 & 0 & 50 \end{bmatrix}$$

uses :

- Shading correction
- Masking (or) Region of Interest (ROI) operations

Division :

0-255

$$\underset{A}{\begin{bmatrix} 0 & 100 & 10 \\ 4 & 0 & 10 \\ 8 & 0 & 5 \end{bmatrix}} \div \underset{B}{\begin{bmatrix} 10 & 100 & 5 \\ 2 & 0 & 0 \\ 0 & 10 & 10 \end{bmatrix}} = \begin{bmatrix} 0 & 1 & 2 \\ 2 & 0 & 0 \\ 0 & 0 & 0.5=0 \\ & & \text{Round off} \end{bmatrix}$$

uses

- Shading correction.

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Logical operations on Images

(i) AND

Truth table

A	B	y
0	0	0
0	1	0
1	0	0
1	1	1

$$X = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} \quad Y = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$X \text{ AND } Y = Z = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(ii) OR

Truth table

A	B	y
0	0	0
0	1	1
1	0	1
1	1	1

$$X = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} \quad Y = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$X \text{ OR } Y = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

(iii) NOT

Truth table

A	B
0	1
1	0

$$X = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} \quad Y = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$X \text{ NOT } Y$$

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[Digital/Image negative, thresholding, clipping, Bit-plane]

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point Operations and its method

They are a method of image processing in which each pixel in the output image is only dependent upon the corresponding pixel in the input image and is independent of its location on neighbouring pixels.

Let 'r' be the gray value at a point (x, y) of the input image $f(x,y)$ and 's' be the gray value at a point (x,y) of the output image $g(x,y)$, then the point operation can be defined as:

$$S = T(r)$$

where, T is the point operation of a certain gray level mapping relationship btn the original image and the output image.

Types of point operations

Intensity value [0 to L-1]

(1) Digital negative / Image negative

$S = (L-1) - r$

$$\begin{bmatrix} 4 & 3 & 5 & 2 \\ 3 & 6 & 4 & 6 \\ 2 & 2 & 6 & 5 \\ 7 & 6 & 4 & 1 \end{bmatrix} \rightarrow \text{input images}$$

(3) = 8 ↓ L

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Assuming the image to be 3 bit, we have $2^3 = 8$ levels.

number of bits, $n = 3$

$$L = 2^n = 2^3 = 8$$

$r = 0$, $s = (8-1) - 0 = 7$

$r = 1$, $s = 7 - 1 = 6$

$r = 2$, $s = 7 - 2 = 5$

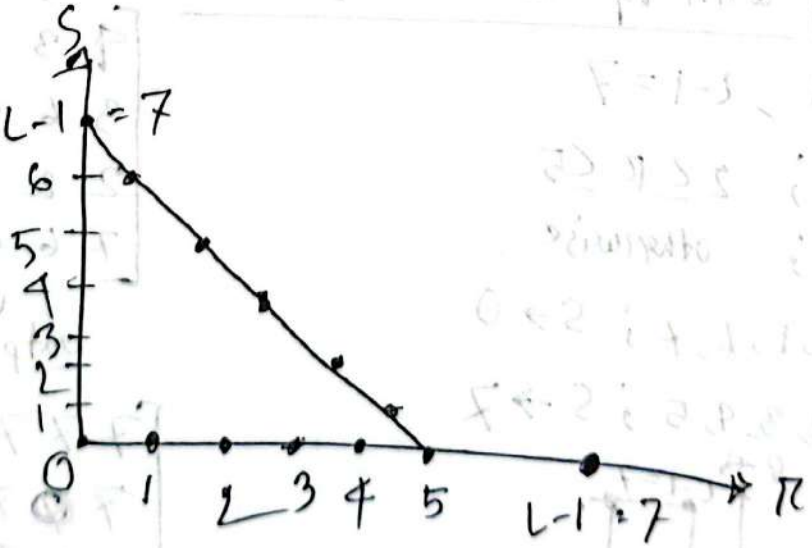
⋮ ⋮ ⋮

$r = 7$, $s = 7 - 7 = 0$ (Highest internal value 7) [0 to 7]

intensity le

output images

$$\begin{bmatrix} 7-4 & 7-3 & 7-5 & 7-2 \\ 7-3 & 7-6 & 7-4 & 7-6 \\ 7-2 & 7-2 & 7-6 & 7-5 \\ 7-7 & 7-6 & 7-4 & 7-1 \end{bmatrix} = \begin{bmatrix} 3 & 4 & 2 & 5 \\ 4 & 1 & 3 & 1 \\ 5 & 5 & 1 & 2 \\ 0 & 1 & 3 & 6 \end{bmatrix}$$



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② Thresholding with $T = 4$

input image

$$\begin{bmatrix} 4 & 3 & 5 & 2 \\ 3 & 6 & 4 & 6 \\ 2 & 2 & 6 & 5 \\ 7 & 6 & 4 & 1 \end{bmatrix}$$

$L = 8$ $(n = 3)$

$L-1 = 7$

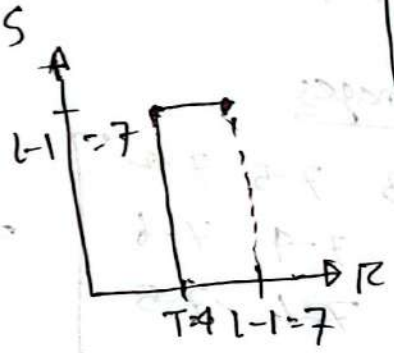
$$s = \begin{cases} L-1 = 7 & ; r \geq 4 \\ 0 & ; r < 4 \end{cases}$$

$r = 0, 1, 2, 3$; $s = 0$

$r = 4, 5, 6, 7$; $s = 7$

output image

$$\begin{bmatrix} 7 & 0 & 7 & 0 \\ 0 & 7 & 7 & 7 \\ 0 & 0 & 7 & 7 \\ 7 & 7 & 7 & 0 \end{bmatrix}$$



③ Clipping with $r_1 = 2$ and $r_2 = 5$

input image

$$\begin{bmatrix} 4 & 3 & 5 & 2 \\ 3 & 6 & 4 & 6 \\ 2 & 2 & 6 & 5 \\ 7 & 6 & 4 & 1 \end{bmatrix}$$

$L = 8$, $L-1 = 7$

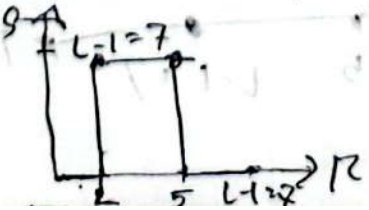
$$s = \begin{cases} L-1 ; & 2 \leq r \leq 5 \\ 0 ; & \text{otherwise} \end{cases}$$

$r \rightarrow 0, 1, 6, 7$; $s \rightarrow 0$

$r \rightarrow 2, 3, 4, 5$; $s \rightarrow 7$

output image

$$\begin{bmatrix} 7 & 7 & 7 & 7 \\ 7 & 7 & 7 & 7 \\ 7 & 7 & 7 & 7 \\ 7 & 7 & 7 & 0 \end{bmatrix}$$



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④ Bit-plane Slicing

input image

$$\begin{bmatrix} 4 & 3 & 5 & 2 \\ 3 & 6 & 4 & 6 \\ 2 & 2 & 6 & 5 \\ 7 & 6 & 4 & 1 \end{bmatrix}$$

$n = 3$

$L = 2^n = 2^3 = 8$

3-bit plane



0 → 000
1 → 001
2 → 010
3 → 011
4 → 100
5 → 101
6 → 110
7 → 111

all are convert binary using 3-bit ($n=3$)

output image

$$\begin{bmatrix} 100 & 011 & 101 & 010 \\ 011 & 110 & 100 & 110 \\ 010 & 010 & 110 & 101 \\ 111 & 110 & 100 & 001 \end{bmatrix}$$

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→ 1 Dimensional fourier transform

$$F(u) = \int_{-\infty}^{\infty} f(x)\, e^{-j2\pi ux}\, dx$$

Inverse ← $$f(x) = f^{-1}(u) = \int_{-\infty}^{\infty} F(u)\, e^{j2\pi ux}\, dx$$

$$|F(u)| = \sqrt{R^2 + I^2}$$

3.3 → Anand

3.8 → Properties

2 Dimensional FT

$$f(u,v) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\, e^{-j2\pi(x+y)u}\, dx\, dy$$

3 Dimen DFT

$$f(u,v,w) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y,z)\, e^{-j2\pi(x+y+z)u}\, dx\, dy\, dz$$

Discrete DFT (N^2) — Times

$$F(u) = \frac{1}{N}\sum_{i=0}^{N-1} f(x)\, e^{-j2\pi ux/N}$$

[N ಸಮಾನ ಭಾಗ]

where, $u = 0, 1, \ldots N-1$

Inverse ← $$f(x) = \frac{1}{N}\sum_{u=0}^{N-1} F(u)\, e^{j2\pi ux/N}$$

$$\therefore F(u,v) = \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)\, e^{-j2\pi(ux+vy)/N}$$

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properties of DFT :

(i) separability

(ii) Translation

(iii) periodicity and conjugate symmetry

(iv) Rotation

(v) Distributivity & scaling

(vi) Average value

(vii) Laplacian

(viii) correlation and convolution

FFT ($N \log_2 N$) Times

$$F(u) = \frac{1}{N} \sum_{n=0}^{N-1} f(x)\, \omega_N^{ux} \quad ; \quad \omega_N = e^{-j\frac{2\pi}{N}}$$

DCT → Example

next Sunday → CT → convolution, Histogram correlation

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Discrete cosine Transform (DCT)

Image compression

(*) One dimensional DCT

$$c(u) = \alpha(u) \sum_{x=0}^{N-1} f(x) \cos \frac{(2x+1)u\pi}{2N}$$

$$\alpha(u) = \begin{cases} \sqrt{\frac{1}{N}} & \text{for } u = 0 \\ \sqrt{\frac{2}{N}} & \text{for } u = 1, 2, 3, \ldots, N-1 \end{cases}$$

(*) Two dimensional DCT can be defined as:

$$c(u,v) = \alpha(u)\alpha(v) \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x,y) \cos\left[\frac{(2x+1)u\pi}{2N}\right] \cos\left[\frac{(2y+1)v\pi}{2N}\right]$$

(*) compute 2D DCT for the following image.

$$f(x,y) = \begin{bmatrix} 2 & 6 \\ 4 & 4 \end{bmatrix}$$

[eqn of c(u,v) written previous line]

$$f(x,y) = \begin{bmatrix} f(0,0) & f(0,1) \\ f(1,0) & f(1,1) \end{bmatrix} = \begin{bmatrix} 2 & 6 \\ 4 & 4 \end{bmatrix}$$

$$c(u,v) = \begin{bmatrix} c(0,0) & c(0,1) \\ c(1,0) & c(1,1) \end{bmatrix}$$

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N = 2

N − 1 = 1

u = v = 0

$$c(0,0) = \alpha(0)\alpha(0) \sum_{x=0}^{1} \sum_{y=0}^{1} f(x,y) \cos\frac{(2x+1)0.\pi}{2.2} \cos\frac{(2y+1)0.\pi}{2.2}$$

$u=0,\ \alpha(u) = \frac{1}{\sqrt{N}}$

$v=0,\ \alpha(v) = \frac{1}{\sqrt{N}};\ N=2$

$\cos 0^\circ = 1$

$$c(0,0) = \sqrt{\frac{1}{2}} \cdot \sqrt{\frac{1}{2}} \sum_{x=0}^{1} \sum_{y=0}^{1} f(x,y) \cdot 1 \cdot 1$$

$$= \frac{1}{2}[f(0,0) + f(0,1) + f(1,0) + f(1,1)]$$

$$= \frac{1}{2}(2+6+4+4)$$

$$= 8 \text{ [value of origin]}$$

$u=0,\ v=1$ for

$$c(0,1) = \alpha(0)\alpha(1) \sum_{x=0}^{1} \sum_{y=0}^{1} f(x,y) \cos\frac{(2x+1)\cdot 0.\pi}{2.2} \cos\frac{(2y+1)1.\pi}{2.2}$$

$$= \sqrt{\frac{1}{2}}\sqrt{\frac{2}{2}} \sum_{x=0}^{1} \sum_{y=0}^{1} f(x,y) \cdot 1 \cos\frac{(2x+1)\pi}{4}$$

$$= \sqrt{\frac{1}{2}} \sum_{x=0}^{1} \sum_{y=0}^{1} [f(0,0)\times\cos\tfrac{\pi}{4} + f(0,1)\times\cos\tfrac{\pi}{4} + f(1,0)\times\cos\tfrac{3\pi}{4} + f(1,1)\times\cos\tfrac{3\pi}{4}]$$

$$= \sqrt{\frac{1}{2}}\left[2\times\frac{1}{\sqrt{2}} + 6\times\frac{1}{\sqrt{2}} + 4\times\left(\frac{-1}{\sqrt{2}}\right) + 4\times\left(\frac{-1}{\sqrt{2}}\right)\right]$$

$$= \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \times (\sqrt{2} + 3\sqrt{2} - 2\sqrt{2} - 2\sqrt{2})$$

$$= \frac{1}{2} \times 6 = \frac{1}{\sqrt{2}}$$

$$= 8$$

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Kernel, K

$$K = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

1D DCT = [K][F]

2D DCT = [K][F][K]^T

Kernel of a four-point DCT

$$\begin{bmatrix} 0.5 & 0.5 & 0.5 & 0.5 \\ 0.6532 & 0.2706 & -0.2706 & -0.6532 \\ 0.5 & -0.5 & -0.5 & 0.5 \\ 0.2706 & -0.6532 & 0.6532 & -0.2706 \end{bmatrix}$$

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The HAAR Transform :

$$h_0(z) \triangleq h_{00}(z) = \frac{1}{\sqrt{N}} \quad \text{for } z \in (0,1)$$

$$= \frac{1}{\sqrt{N}} \begin{cases} 2 \cdot \frac{p}{2} & \frac{q-1}{2p} \le z \le \frac{q-\frac{1}{2}}{2p} \\ -2 \cdot \frac{p}{2} & \frac{q-\frac{1}{2}}{2p} \le z \le \frac{q}{2p} \end{cases}$$

2×2 matrices or Kernel

↳ $A_2 = \frac{1}{\sqrt{2}} \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}$

4×4 matrices or Kernel

↳ $A_4 = \frac{1}{\sqrt{4}} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ -\sqrt{2} & -\sqrt{2} & 0 & 0 \\ 0 & 0 & -\sqrt{2} & -\sqrt{2} \end{pmatrix}$

Question :

- what is FT ?
- what is role in IP ?
- what do you mean F spectrum & F power spectrum ?
- write the eqn for DFT of 1 & 2 Dimensional fun for images ?
- compute the F
- find the F spectrum for the fun $f(x) = \{4, 3, 2, 1\}$ for $x = 0, 1, 2, 3$
- State the convolution & correlation theorem ?

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: State the use of FT & DCT ?

→ New image generate by convolution & HAAR Transform practices.

. 2 Dimensional system practices.

→ Compute the new image from the following image using DCT and HAAR Transform ?

Lab-05

(i) DCT
(ii) FFT
(iii) HAAR Transform using matrix

Lab-06

(i) High pass filter
(ii) convolution
(iii) correlation
(iv) Histogram equilization.

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Haar transform :

given the matrix is -

$$A = \begin{bmatrix} 52 & 55 & 61 & 66 \\ 70 & 61 & 64 & 73 \\ 63 & 65 & 66 & 68 \\ 67 & 70 & 70 & 75 \end{bmatrix}$$

Row-wise Haar :

Apply on each row :

(52, 55, 61, 66)

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pairs :

$$\text{Avg} = \frac{52+55}{2} = 53.5$$

$$\text{Diff} = \frac{52-55}{2} = -1.5$$

$$\text{Avg} = \frac{61+66}{2} = 63.5$$

$$\text{Diff} = \frac{61-66}{2} = -2.5$$

Row Result = (53.5, 63.5, -1.5, -2.5)

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After all rows:

$$\begin{bmatrix} 53.5 & 63.5 & -1.5 & -2.5 \\ 65.5 & 68.5 & 4.5 & -4.5 \\ 64 & 67 & -1 & -1 \\ 68.5 & 72.5 & -1.5 & -2.5 \end{bmatrix}$$

Column-wise Haar:

apply same idea column wise:

Take first column:

$(53.5, 65.5, 64, 68.5)$

pairs:

Avg: $\frac{53.5+65.5}{2} = 59.5$

Diff: $\frac{53.5-65.5}{2} = -6$

Avg: $\frac{64+68.5}{2} = 66.25$

Diff: $\frac{64-68.5}{2} = -2.25$

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final Haar Transform :

$$H(A) = \begin{bmatrix} -59.5 & 66 & -1 & -2.5 \\ 66.25 & 69.75 & -1.25 & -1.75 \\ -6 & -2.5 & 3 & -1 \\ -2.25 & -2.75 & 0.25 & -0.75 \end{bmatrix}$$ (Ans)

(*) DCT (Discrete Cosine Transform)

For a 4×4 image

$$F = K \times A \times K^T$$

$$K = \begin{bmatrix} 0.5 & 0.5 & 0.5 & 0.5 \\ 0.65 & 0.27 & -0.27 & -0.65 \\ 0.5 & -0.5 & -0.5 & 0.5 \\ 0.27 & -0.65 & 0.65 & -0.27 \end{bmatrix}$$

$$1D\ DCT = K \times A$$

$$= \begin{bmatrix} 0.5 & 0.5 & 0.5 & 0.5 \\ 0.65 & 0.27 & -0.27 & -0.65 \\ 0.5 & -0.5 & -0.5 & 0.5 \\ 0.27 & -0.65 & 0.65 & -0.27 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 1 & 1 \\ 5 & 2 & 5 & 6 \\ 3 & 3 & 2 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

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$$2\begin{bmatrix} 3.5 & 4 & 4 & 5 \\ -0.85 & 0.19 & 0.08 & 0.85 \\ -2.5 & -1 & 2 & 2 \\ 0.75 & 1.12 & -2.04 & -2.58 \end{bmatrix}$$

2D DCT $= \begin{bmatrix} 3.5 & 4 & 4 & 5 \\ -0.85 & 0.19 & 0.08 & 0.85 \\ -2.5 & -1 & 2 & 2 \\ 0.75 & 1.12 & -2.04 & -2.58 \end{bmatrix}$

$$\times \begin{bmatrix} 0.5 & 0.65 & 0.5 & 0.27 \\ 0.5 & 0.27 & -0.5 & -0.65 \\ 0.5 & -0.27 & -0.5 & -0.65 \\ 0.5 & -0.65 & 0.5 & -0.27 \end{bmatrix}$$

$$= \begin{bmatrix} 8.25 & -1.73 & -0.75 & 0.48 \\ -0.73 & -1.03 & -1.42 & 0.82 \\ -0.75 & -1.25 & -1.25 & 0.75 \\ -1.12 & 0.53 & -0.08 & 0.53 \end{bmatrix}$$

(Ans)

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Name: Tushare 11-2-2026

TOPIC NAME: DIP- CT-6403 (Assignment) DAY: TIME: DATE: / /

⑧ Histogram

CT

. what is histogram ?

→ A histogram is a graphical representation of the distribution of pixel intensity values (gray scales) in an image. It shows how many pixels in the image have possible intensity value.

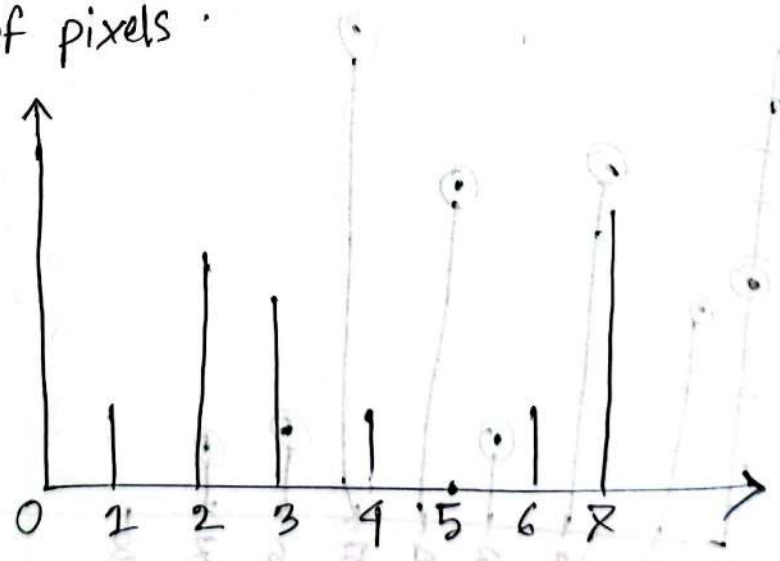
For an image, the histogram is defined as:

$$h(r_k) = n_k$$

where,

r_k = Intensity (gray) level

n_k = Number of pixels having intensity r_k.



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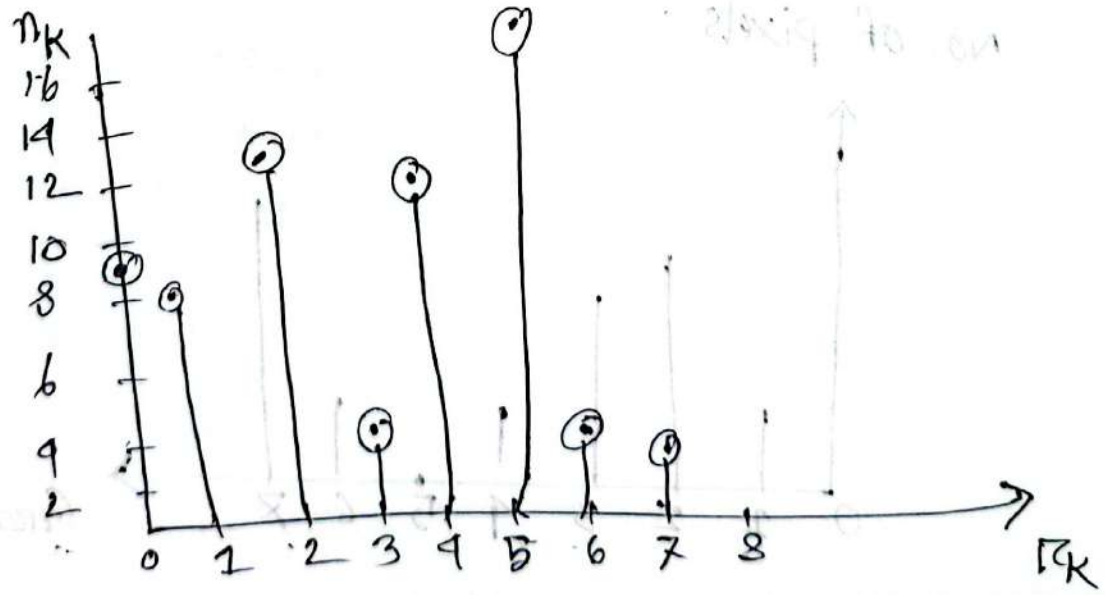
⊛ Application of histogram :

- Image enhancement
- Histogram equalization
- Image thresholding
- Image segmentation.
- Image compression.

Spatial Domain

Q-01: perform the histogram equalization for an input image shown below.

Gray levels	0	1	2	3	4	5	6	7
No. of pixels	9	8	11	4	10	15	4	3



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$$S_k = n_0 K + n_1 K$$

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Histogram of input image

Gray levels	no of pixels	$P(r_k) = \frac{n_k}{n}$	S_k (CDF)	$S_k \times 7$	Histogram equalization
0	9	0.141	0.141	0.987	1
1	8	0.125	0.266	1.862	2
2	11	0.172	0.438	3.066	3
3	4	0.0625	0.5005	3.5035	4
4	10	0.156	0.6565	4.5955	5
5	15	0.234	0.8905	6.2335	6
6	4	0.0625	0.953	6.671	7
7	3	0.0469	1	7	7
	n = 64				

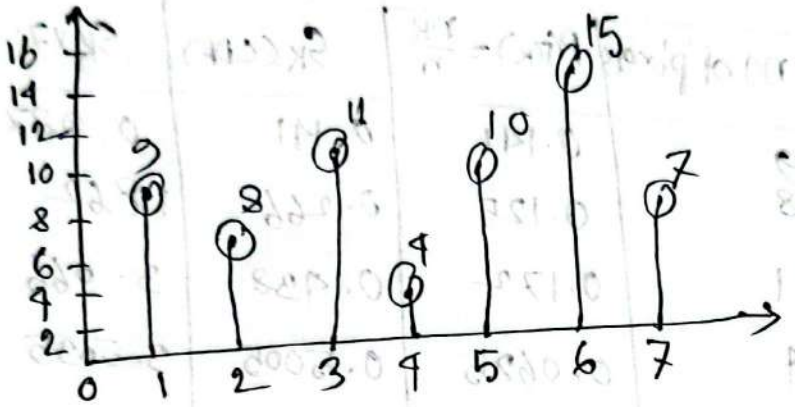
Gray levels	No. of pixels
1	9
2	8
3	11
4	4
5	10
6	15
7	7

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No. of pixel (n_k)



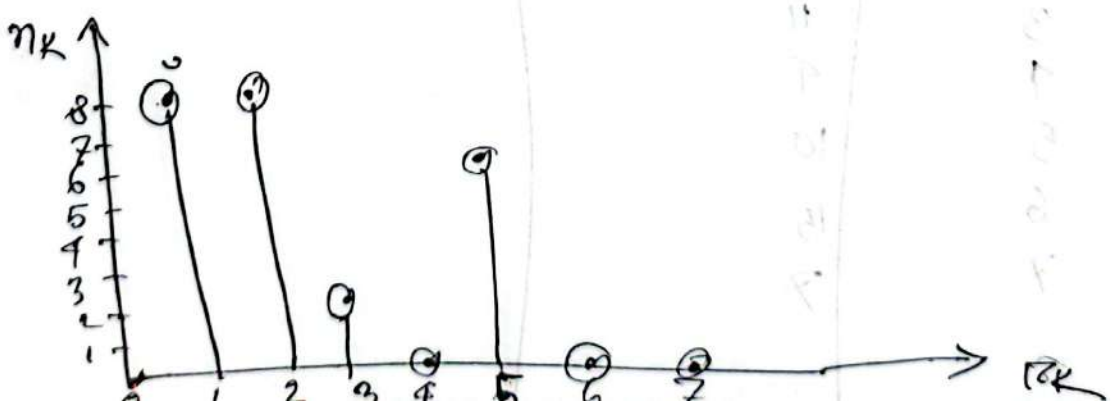
Q-02: perform histogram equalization for the folbwing image: $f(x,y)$ =

1	2	1	1	1
2	5	3	5	2
2	5	5	5	2
2	5	3	5	2
1	1	1	2	1

max value = 5

Soln:

Gray levels (r_k)	0	1	2	3	4	5	6	7
No. of pixels (n_k)	0	8	8	2	0	7	0	0

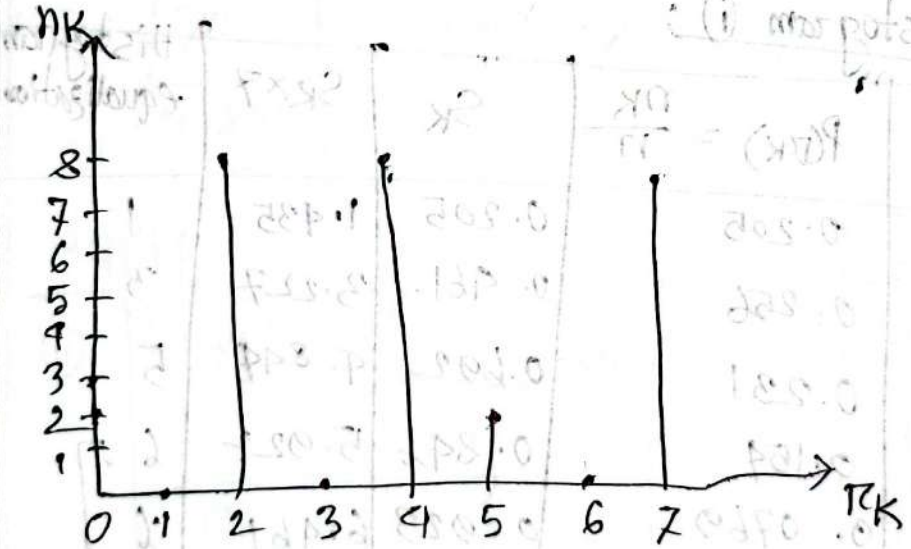


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Gray level	No. of Pixels	$P(r_k) = \frac{n_k}{n}$	SK(CDF)	SK x 7	Histogram equalization
0	0	0	0	0	0
1	8	0.32	0.32	2.24	2
2	8	0.32	0.64	4.48	4
3	2	0.08	0.72	5.04	5
4	0	0	0.72	5.04	5
5	7	0.28	1	7	7
6	0	0	1	7	7
7	0	0	1	7	7
	n=25				



1	2	1	1	1
2	5	3	5	2
2	5	5	5	2
2	5	3	5	2
1	1	1	2	1

Input Image

→

2	4	2	2	2
4	7	5	7	4
4	7	7	7	4
4	7	5	7	4
2	2	2	4	2

output Image

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Histogram Matching

Q→ Given below are two histogram (i) and (ii). Modify the histogram (i) as given by histogram (ii)

(i)

Gray level (r_k)	0	1	2	3	4	5	6	7
No. of pixels (n_k)	80	100	90	60	30	20	10	0

(ii)

Gray level (z_k)	0	1	2	3	4	5	6	7
No. of pixels (n_k)	0	0	0	60	80	100	80	70

Ans

Equalize histogram (i):

Gray level	no. of pixels	$P(r_k) = \frac{n_k}{n}$	Sk	Sk×7	Histogram equalization	New r_k
0	80	0.205	0.205	1.435	1	80
1	100	0.256	0.461	3.227	3	100
2	90	0.231	0.692	4.844	5	90
3	60	0.154	0.846	5.922	6	
4	30	0.0769	0.923	6.461	6	90
5	20	0.0513	0.974	6.818	7	
6	10	0.0256	0.999	6.993	7	80
7	0	0	0.999	6.993	7	

n = 390

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(CT) Equalize histogram (ii)!

Gray level	No. of pixels	$P(r_k) = \frac{n_k}{n}$	S_k	$S_k \times 7$	Histogram equalization
0	0	0	0	0	0
1	0	0	0	0	0
2	0	0	0	0	0
3	60	0.154	0.154	1.078	1
4	80	0.205	0.359	2.513	3
5	100	0.256	0.615	4.305	4
6	80	0.205	0.82	5.74	6
7	70	0.179	1	7	7

$n = 390$

Mapping

(ii)

Gray level	Histogram equalization level
0	0
1	0
2	0
3	1
4	3
5	4
6	6
7	7

(i)

Gray level		Histogram equalization level
0	1	80
1	3	100
2	5	90
3	6	90
4	6	90
5	7	30
6	7	30
7	7	30

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(Q1) **Histogram equalization :-** Histogram equalization is a technique that redistributes the gray levels of an image so that the histogram becomes approximately uniform.

* Why do we use this techniques in image enhancement techniques?

→ Image enhancement techniques are used to improve the visual quality of an image, increase contrast, and make important details more visible. Histogram based methods are widely used for this purpose.

It is used to →

- improve the overall contrast of an image.
- make hidden details more visible
- utilize the full range of available intensity levels.
- enhance images that are too dark or too bright.

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Histogram matching (Specification) :-

Histogram matching is a technique in which the histogram of an input image is transformed to match a specified target histogram.

It is used to →

- make different images appear similar.
- standardize images for analysis and comparison.
- controll the appearance of the enhanced image.

Application :

- Remote sensing
- Medical image processing
- pattern recognition

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(CT)

⊛ Local Histogram Equalization :

Local histogram equalization applies histogram equalization to small regions of an image instead of the entire image.

It is used to →

- improve local contrast.
- reveal details in dark and bright regions simultaneously.
- enhance images with non-uniform illumination.
- preserve local image features.

Application :

- Medical imaging
- Face recognition
- Low-light image enhancement.

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Q → Proof of probability distribution of output image in histogram equalization.

→ $P_S(s) = \frac{1}{L-1}$, $0 \le s \le L-1$

Given, Let,

r = gray level of the input image

$P_r(r)$ = PDF of the image

L = total number of gray levels

s = transformed gray level after histogram equalization.

The transformation function is,

$$s = T(r) = (L-1)\int_0^r P_r(w)\,dw$$

where, $T(r)$ is the cumulative distribution function (CDF).

Differentiating both sides with respect for,

$$\frac{ds}{dr} = (L-1)\,P_r(r)$$

$$\Rightarrow \frac{dr}{ds} = \frac{1}{(L-1)\,P_r(r)}$$

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for monotonic transformation -

$$P_S(s) = P_R(r) \left| \frac{dr}{ds} \right|$$

Substituting $\frac{dr}{ds}$,

$$P_S(s) = P_R(r) \times \frac{1}{(L-1) P_R(r)}$$

$$\Rightarrow P_S(s) = \frac{1}{L-1} \text{ for } 0 \leq s \leq L-1$$

$$\therefore P_S(s) = \frac{1}{L-1}, \quad 0 \leq s \leq L-1$$

(Proved)

(*) Define convolution and correlation:

CT

Convolution:

convolution is a mathematical operation used in digital image processing to combine an image with a filter (kernel) to produce a new image. In convolution the kernel is flipped both horizontally and vertically before multiplication and summation.

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for 2D images -

$$g(x,y) = f(x,y) * h(x,y)$$

$$g(x,y) = \sum_m \sum_n f(m,n) \cdot h(x-m, y-n)$$

Application :

- Image smoothing
- Sharpening
- Edge detection
- Noise reduction

Correlation:

correlation is a mathematical operation used to measure the similarity between an image and a template (Kernel). The Kernel is not flipped before multiplication and summation..

$$g(x,y) = f(x,y) \otimes h(x,y)$$

$$g(x,y) = \sum_m \sum_n f(m,n) \, h(x+m, y+n)$$

Applications:

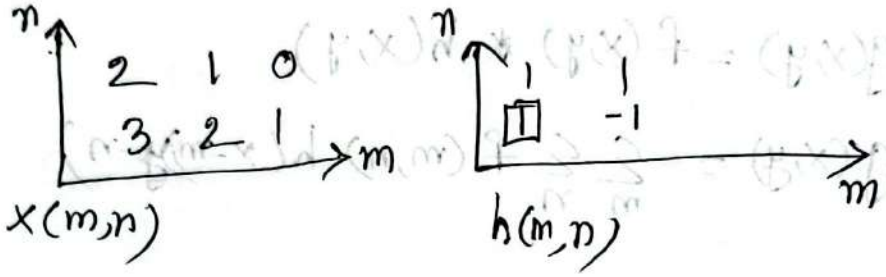
- Template matching
- pattern recognition
- object detection

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Q → Determine the convolution of $x(m,n)$ with $h(m,n)$.



Soln:

Flip the kernel horizontally and vertically.

$h(m,-n)$

1	-1
1	1

$h(-m,-n)$

-1	1
1	1

$$h_f = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

(padding) $$x_p = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 \\ 0 & 3 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

output size = $(2+2-1) \times (3+2-1) = 3 \times 4$

Row 1,

$$y_{11} = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} = 2$$

$$y_{12} = \begin{bmatrix} 0 & 0 \\ 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} = 3$$

$$y_{13} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} = 1$$

"GOOD LUCK"

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$y_{14} = 0$

Row 2,

$$y_{21} = \begin{bmatrix} 0 & 2 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} = 2+3 = 5$$

$$y_{22} = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} = 4$$

$$y_{23} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} = 2$$

$$y_{24} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} = 1$$

Row 3,

$y_{31} = 3$

$y_{32} = 5$

$y_{33} = 3$

$y_{34} = 1$

$\therefore Y = \begin{bmatrix} 2 & 3 & 1 & 0 \\ 5 & 4 & 2 & 1 \\ 3 & 5 & 3 & 1 \end{bmatrix}$ (Ans)

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3×3 2×2 TIME : DATE : / /

CT

Q→

$x(m,n)$

4	5	3
3	2	1
1	0	-1

$h(m,n)$

1	0
0	-1

Soln:

Flip the kernel horizontally and vertically,

$h(m,-n)$

0	-1
1	0

$h(-m,-n)$

-1	0	
0	1	

$$hf = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

Output size = $(3+2-1) \times (3+2-1) = 4 \times 4$

$$x_p = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 4 & 5 & 3 & 0 \\ 0 & 3 & 2 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Row 1,

$$y_{11} = \begin{bmatrix} 0 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = 4$$

$$y_{12} = \begin{bmatrix} 0 & 0 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = 5$$

$$y_{13} = \begin{bmatrix} 0 & 0 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 1 & 0 \end{bmatrix} = 3$$

$$y_{14} = \begin{bmatrix} 0 & 0 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 1 & 0 \end{bmatrix} = 0$$

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Row 2,

$y_{24} = \begin{bmatrix} 0 & 4 \\ 0 & 3 \end{bmatrix} \cdot \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = 3$

$y_{23} = \begin{bmatrix} 5 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = -4$

$y_{22} = \begin{bmatrix} 4 & 5 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = -2$

$y_{29} = \begin{bmatrix} 3 & 0 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = -3$

Row 3,

$y_{31} = \begin{bmatrix} 0 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = 1$

$y_{33} = \begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = -3$

$y_{32} = \begin{bmatrix} 3 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = -3$

$y_{34} = \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = -1$

Row 4,

$y_{41} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = 0$

$y_{43} = \begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = 0$

$y_{42} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = -1$

$y_{44} = \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = 1$

$\therefore Y = \begin{bmatrix} 4 & 5 & 3 & 0 \\ 3 & -2 & -4 & -3 \\ 1 & -3 & -3 & -1 \\ 0 & -1 & 0 & 1 \end{bmatrix}$ (Ans)

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6.2.3 : HSI color Model :

Hue
Saturation
Intensity

Chrominance
Luminance

RGB & CMY → cyan, magenta, yellow

SNR → Signal to Noise ratio

MSE → mean square Error

$$\frac{1}{mean} \sum_{K=0}^{M-1} \sum_{K=0}^{N-1} f(K,K) = f(x,y)$$

$x=0 \quad y=0 \rightarrow f(x,y)$

$$\text{Error} = \underbrace{f(x,y)}_{\text{Input Image}} - \underbrace{f'(x,y)}_{\text{output Image}}$$

$$SNR = \frac{\sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [f(x,y)]^2}{\sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [f(x,y) - f'(x,y)]^2}$$

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TOPIC NAME : MATLAB DAY : _____

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(q) imread() → for built in image nowa hoi]

(r) imnoise () → image-a noise create (gaussian noise)

↳ imnoise (built in image, gaussian, gaussian noise, GN & variance)

(n) im2double () → img data type ke double a ante

OC - 13.07.25

Ray optics (basic laws)

Refraction, optical fiber structure & communication

light > critical angle
and reflect same medium

Lab-01: How to strip of the jacket to expose the cladding. (cleave)

Numerical Aperture, $N_A = n_0 \sin\theta$

$$NA \propto \frac{1}{L_c}$$

Pulse spreading → in single & multimode

coupler → Antenna (CS), LED

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Data compression : reduce data size

Image compression :

2D array reduce without loss array information

Redundency → $R = 1 - \frac{1}{C}$

Compression ratio : $C = \frac{n_1}{n_2}$

Remote sensing
Documentation
Medical Imaging
Video conference

* coding Redundency
spatial & temporary or (interpixel) Redundency. ← corelated info
psychovisual or irrevalent information redundency
Info - ↳ when
but our human

* pixel value → 0-255 , Imge size = 256 × 256
Num of bits → 256 × 256 × 8 (storage size of image)

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Coding Redundancy :

r_k	$P_r(r_k)$	code	$l(r_k)$
$r_{87} = 87$	0.25	01	2
$r_{128} = 128$	0.47	1	1
$r_{186} = 186$	0.25	000	3
$r_{255} = 255$	0.03	001	3

$$L_{avg} = \sum_{k=0}^{L-1} l(r_k)\, P_r(r_k)$$

$r_k \rightarrow$ intensity values

$l(r_k)$ = No. of bits used to represent r_k.

$L_{avg} = 2 \times 0.25 + 0.47 \times 1 + 0.25 \times 3 + 0.03 \times 3$

$= 1.81$ bits

$$C = \frac{256 \times 256 \times 8}{256 \times 256 \times 1.81} = 4.42$$

$$\therefore R = 1 - \frac{1}{C} = 1 - \frac{1}{4.42} = 0.774$$

$= 77.4\%$

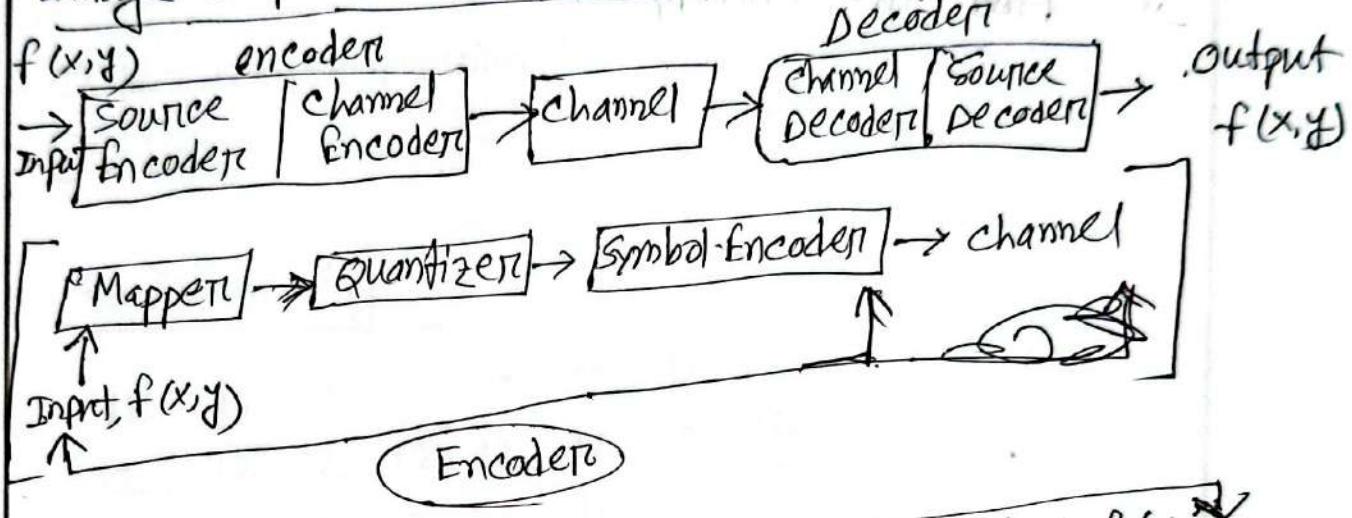
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(Anvel book)

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Image Compression Models:



Decoder

channel → symbol decoder → Inverse Mapper → output, f(x,y)

Huffman code

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DATE : / /

Q! find the entropy of

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① Lossless compression

② Lossy "

Huffman coding:

12 12 12 12 12 12 12 12

12 12 12 12 12 12 12 12

12 12 12 12 12 12 12 12

12 12 200 200 200 100 100 100

~~50~~ ~~50~~ 200 200 200 100 100 100

50 50 200 200 200 100 100 100

50 50 20 20 20 5 5 5

50 50 20 20 20 5 5 5

Gray level : $8 \times 8 \rightarrow 64$

$P_1 = 5 \rightarrow 6 = \frac{6}{64} = 0.094$

$P_2 = 12 \rightarrow 26 = \frac{26}{64} = 0.406$

$P_3 = 20 \rightarrow 6 = \frac{6}{64} = 0.094$

$P_4 = 50 \rightarrow 8 = \frac{8}{64} = 0.125$

$P_5 = 100 \rightarrow 9 = \frac{9}{64} = 0.141$

$P_6 = 200 \rightarrow 9 = \frac{9}{64} = 0.141$

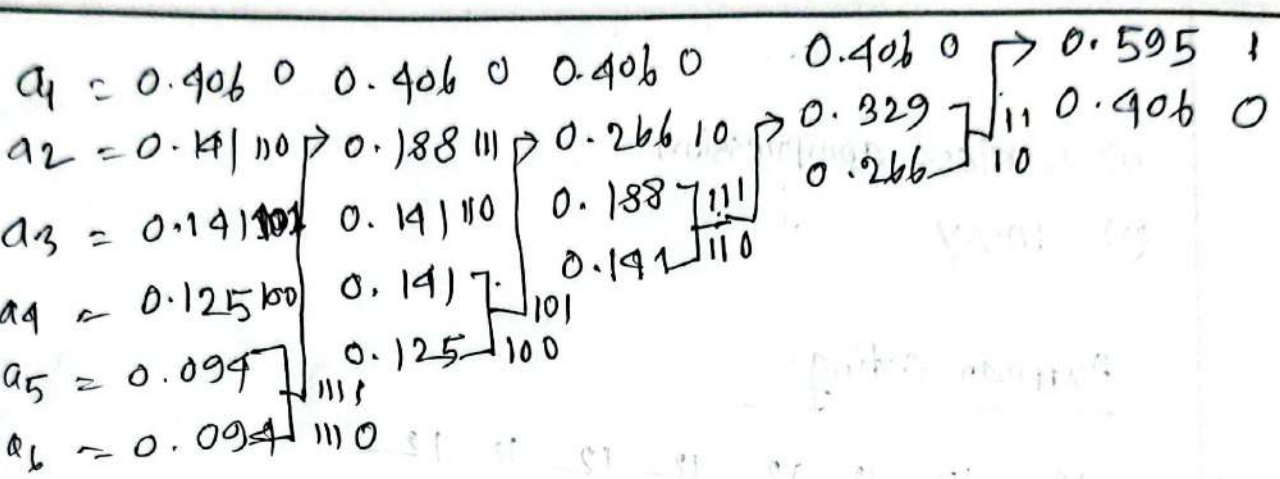
H =

Descending order → 0.406 0.141 0.141 0.125 0.094 0.094

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$L_{avg} \rightarrow 0.406 \times 1 + 0.141 \times 3 + 0.141 \times 3 + 0.125 \times 3$

$+ 0.094 \times 4 + 0.094 \times 4$

$= 0.406 + 0.423 + 0.423 + 0.375 + 0.376$

$+ 0.376$

$= 2.379$

Total bits = $64 \times 8 = 512$

Needed bits = $64 \times 2.379 = 152.256$

compression ratio =

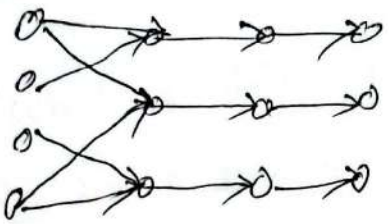
GOOD LUCK

TOPIC NAME : DIP DAY : TIME : DATE : / /

5.15: Image compression using neural network.

Multilayer perceptron network for image.

Back propagation network for image compression.
↳ methodology



x_i (input) -> y_i (output)

wave, w_i

$$y = f\left[\sum_{i=0}^{n-1} w_i x_i\right]$$

PSNR

MSE -> (?) -> x

compression ratio

[CR↑ sys perfo ↑]

Ques: (i) compute the entropy

(ii) construct the huffman code calculation

(iii) Avg bit calculation

(iv) what is CR? what are the types of redundancy? Briefly express the difference of redundancy?

(v) Explain the

(vi) what is entropy?

(vii) State & explain the fun' of source encoder & decoder

(viii) lossy vs lossless compression

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Image segmentation

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CT-05 (Image compression)

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Image compression: In DIP, Image compression is the process of reducing the amount of data required to represent a digital image. (websites, medical imaging, Remote sensing, mobile & digital cameras applications)

Ques: what is data redundency in image compression?

Ans: Data redundency occurs when an image contains superfluous/unnecessary data that can be removed without information. It is measured by, $R = 1 - \frac{1}{C}$

where, C is compression ratio, $C = \frac{n_1}{n_2}$ (original / compressed)

$n_2 < n_1, C = $

$n_2 >> n_1, C = $

Ques-02: Explain the 3 main types of image redundencies?

Ans: Coding Redundency: occurs when gray levels are represented using more code words than necessary.

avg num of bits = b_{avg} or l_{avg}

k^{th} gray level num of bits = $b(g_k)$ or $l(g_k)$ or $l(r_k)$

g_k or r_k = intensity values

kth gray level probability = $p(r_k)$ or $p(g_k)$

$$\therefore l_{avg} = \sum_{k=0}^{L-1} l(r_k) \cdot P_n(r_k)$$

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Example: fixed & variable length codes example

r_k	$P(r_k)$	code 1	$l_1(r_k)$	code 2	$l_2(r_k)$
$r_0 = 0$	0.12	000	3	11	2
$r_1 = 1/7$	0.22	001	3	01	2
$r_2 = 2/7$	0.18	010	3	10	2
$r_3 = 3/7$	0.16	011	3	001	3
$r_4 = 4/7$	0.1	100	3	0001	4
$r_5 = 5/7$	0.2	101	3	00001	5
$r_6 = 6/7$	0.01	110	3	000001	6
$r_7 = 1$	0.01	111	3	0000001	7

code 1: $L_{avg} = \sum_{k=0}^{7} l_1(r_k) \cdot P(r_k)$ $(L = 8)$

$= 3 \times 0.12 + 3 \times 0.22 + 3 \times 0.18 + 3 \times 0.16 + 3 \times 0.1$
$+ 3 \times 0.2 + 3 \times 0.01 + 3 \times 0.01 = 3$ bits

code 2: $L_{avg} = \sum_{k=0}^{7} l_2(r_k) \cdot P(r_k)$

$= 2 \times 0.12 + 2 \times 0.22 + 2 \times 0.18 + 3 \times 0.16 + 4 \times 0.1$
$+ 5 \times 0.2 + 6 \times 0.01 + 7 \times 0.01 = 2.114$ bits

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Spatial / Interpixel Redundency: Occurs due to high correlation between adjacent pixels in an image.

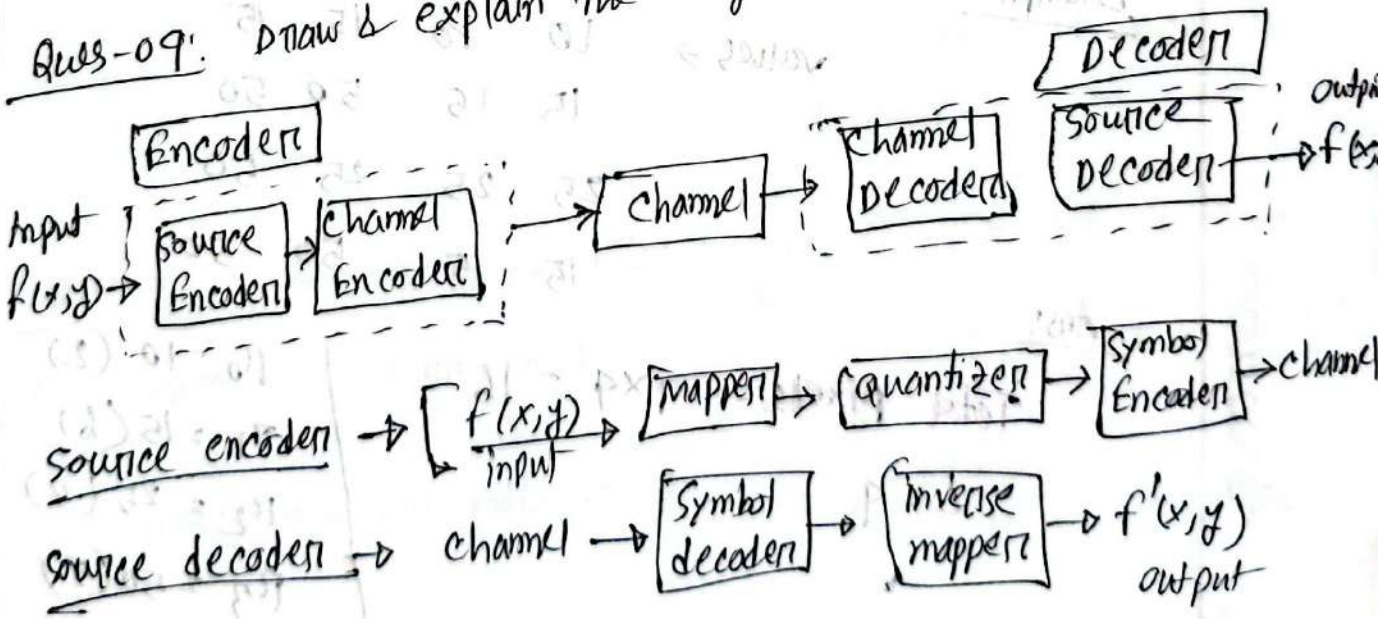
psychovisual or Irrevalent information Redundency: Occurs because the human visual system (HVS) ignores certain visual information.

Ques-03: what is the main difference between lossless and lossy compression?

Ans: Lossless compression allows exact reconstruction of the original image without any data loss. (CR = 2:1 to 3:1)

Lossy compression achieves much higher compression ratios (CR > 10:1) by permanantly discarding psychovisually redundant data.

Ques-04: Draw & explain the image compression model



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Ques-05: Define Entropy (H) of an image?

Ans: Entropy is the average information content generated from all the pixels in image.

Entropy, $H = -\sum_{K=0}^{L-1} P_K \log_2 P_K$ bits/symbol or message.

$= \sum_{K=0}^{L-1} \frac{P_K}{\log_2 P_K}$ $\left[P_K = \frac{1}{L-1}\right]$

$[K = 0, 1, 2, \cdots, L-1]$

$P_K = \frac{\text{rk}}{\text{total pixels}}$

$H = -\sum_{K=0}^{L-1} \frac{1}{L-1} \log_2 \frac{1}{L-1}$

$\max H = \sum_{K=0}^{L-1} \log_2 (L-1)$ bits

P_K = probability
L = intensity/gray levels
r_K = gray levels how time repeat

Example. Find entropy 4×4 image size with gray level

values → 10 10 15 15
15 15 50 50
25 25 25 50
15 15 50 50

Ans:

Total pixels = 4×4 = 16

$L = 4$

$r_0 = 10$ (2)
$r_1 = 15$ (6)
$r_2 = 25$ (3)
$r_3 = 50$ (5)

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Entropy, $H = -\sum_{K=0}^{L-1} P_K \log_2 P_K$ | $P_K = \frac{\text{no. of pixels}}{\text{total pixels}}$

$= -\sum_{K=0}^{q-1} P_K \log_2 P_K$

$= -(P_0 \log_2 P_0 + P_1 \log_2 P_1 + P_2 \log_2 P_2 + P_3 \log_2 P_3)$

$= -\left(\frac{2}{16}\log_2\frac{2}{16} + \frac{6}{16}\log_2\frac{6}{16} + \frac{3}{16}\log_2\frac{3}{16} + \frac{5}{16}\log_2\frac{5}{16}\right)$

$= 1.88$ bits

total $= 1.88 \times 16 = 30.08$ bits

Ques-06: Define compression Ratio (CR)?

Ans: $C_R = \frac{n_1}{n_2}$, where n_1 is the number of bits in original image and

n_2 is the number of bits in compressed image.

Ques-07: write the mathematical formulas for MSE & PSNR?

Ans: mean square Error, $MSE = \frac{1}{mn}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1}[f(x,y) - f(x,y)]^2$

peak signal to noise Ratio, $PSNR = 10 \log_{10}\left(\frac{(2^n-1)^2}{MSE}\right)$ dB

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Gray level → how much time repeats

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Ques-08) Define Huffman coding & give example & coding efficiency?

Ans:

Huffman coding: Is the simplest process that used to compress the image data without any loss is variable length coding.

Example:

Image size = 8×8 = 64

gray level pixes values →

12	12	12	12	12	12	12	12
12	12	12	12	12	12	12	12
12	12	12	12	12	12	12	12
12	12	200	200	200	100	100	100
50	50	200	200	200	100	100	100
50	50	200	200	200	100	100	100
50	50	20	20	20	5	5	5
50	50	20	20	20	5	5	5

Gray level, r_k	probability, P_r
r_0 = 5 (6 times)	6/64 = 0.094
r_1 = 12 (26 times)	26/64 = 0.406
r_2 = 20 (6 times)	6/64 = 0.094
r_3 = 50 (8 times)	8/64 = 0.125
r_4 = 100 (9 times)	9/64 = 0.141
r_5 = 200 (9 times)	9/64 = 0.141

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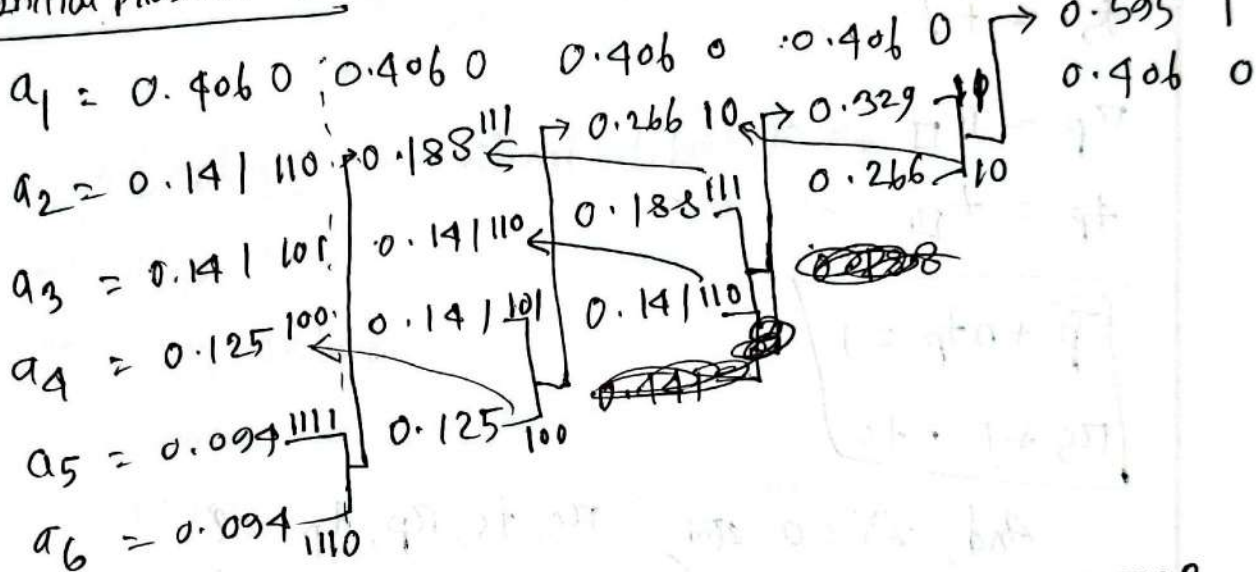
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gray levels are arranged in descending order,

0.406 0.141 0.141 0.125 0.094 0.094

Initial probabilities



$L_{avg} = 0.406 \times 1 + 0.141 \times 3 + 0.141 \times 3 + 0.125 \times 3 + 0.094 \times 4 + 0.094 \times 4$

$= 2.37$ bits/symbol or message

coding efficiency, $\eta = \frac{H \text{ (entropy)}}{L_{avg}} \times 100\%$ (if $H = 2.05$)

$= \frac{2.05}{2.37} \times 100\%$

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Fresnel Eqn.

$$r_s = r_\perp = \frac{E_{r0,\perp}}{E_{i0,\perp}}$$

$$t_s = t_\perp$$

$$r_p = r_\parallel$$

$$t_p = t_\parallel$$

$$\boxed{r_p + n t_p = 1}$$

$$\boxed{r_s + 1 = t_s}$$

And, $\theta_i = 0$ then, $r_s, t_s, r_p, t_p = ?$

when, $r_p = 0$, $\tan\theta_p = n = \frac{n_2}{n_1}$ [θ_p = Brewster angle]

$(n^2 - \sin^2\theta_p)^{1/2}$

critical angle, $\theta_c = \sin^{-1}\frac{n_2}{n_1}$

Brewster angle, $\theta_p = \tan^{-1}\frac{n_2}{n_1}$

(8) Antireflection coating (31) (69)

(i) when $\theta_i = 0$ then 1% reflectance & 1% transmittance

(ii)

$$K_c = \frac{2\pi}{\lambda_c}, \quad \text{path diff} = K_c(2d) = \frac{2\pi \cdot 2\phi}{\lambda_c}$$

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